

DATA-260

HOMEWORK 1

Name: Kshitija Shinde
SJSU ID: 019146491

Value	Definition
SID4	6491
PORT_BASE	$8000 + (6491 \bmod 900) = 8000 + 191 = \mathbf{8191}$
PREFIX	"s" + 6491 = s6491
SEED	6491
VERIFY_SEED	$260000 + 6491 = \mathbf{266491}$
DOMAIN_ID	$6491 \bmod 8 = \mathbf{3}$
Assigned Domain	Grocery supply and recall notices

Hardware: MacBook Air, 13-inch, M4, 2025, Apple M4 chip, 16GB RAM, macOS Tahoe 26.5.2

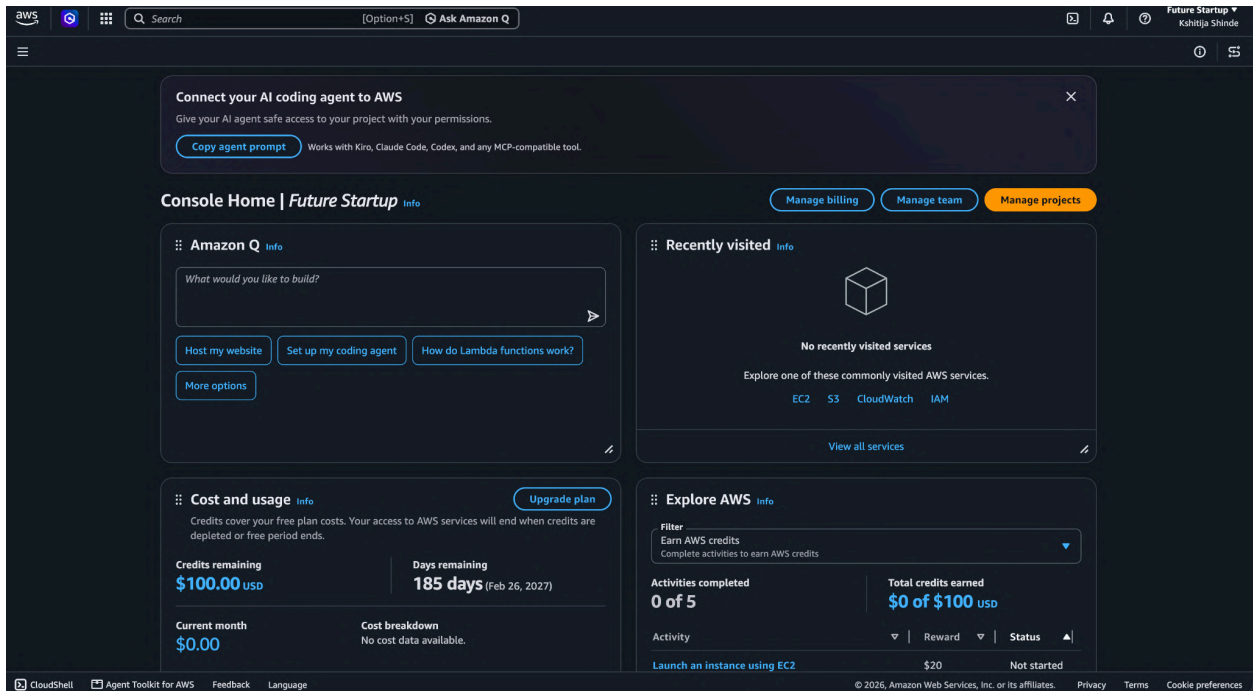
Local model used: qwen3:8b

Tagged commit hash: 3639106ba7b15e0ab54ea5bf80a6abac7a23e01a

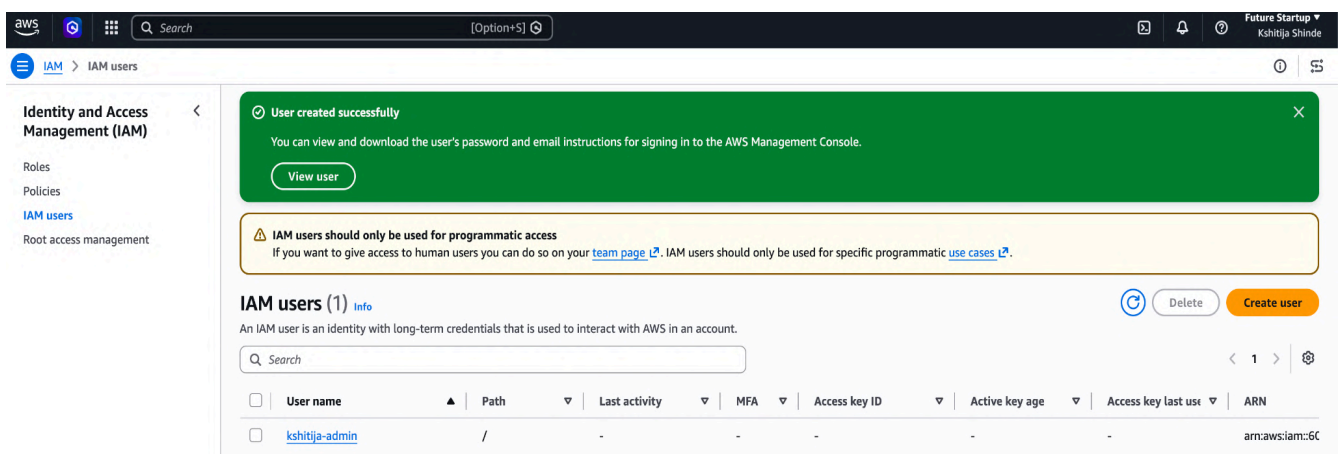
Repository: <https://github.com/Kshitija-Shinde9/data260-6491>

Before coming to the part 1 and further homework here is the setup I did

1. Created AWS account



2. I created IAM user with AdministratorAccess just for security purpose so that AWS CLI can safely talk to my AWS account without putting my entire account at risk.



3. Then created access key

Step 1

Access key best practices & alternatives

Step 2 - optional

Set description tag

Step 3

Retrieve access keys

Access key best practices & alternatives

Avoid using long-term credentials like access keys to improve your security. Consider the following use cases and alternatives.

Use case

☒ Command Line Interface (CLI)

You plan to use this access key to enable the AWS CLI to access your AWS account.

☐ Local code

You plan to use this access key to enable application code in a local development environment to access your AWS account.

☐ Application running on an AWS compute service

You plan to use this access key to enable application code running on an AWS compute service like Amazon EC2, Amazon ECS, or AWS Lambda to access your AWS account.

☐ Third-party service

You plan to use this access key to enable access for a third-party application or service that monitors or manages your AWS resources.

☐ Application running outside AWS

You plan to use this access key to authenticate workloads running in your data center or other infrastructure outside of AWS that needs to access your AWS resources.

☐ Other

Your use case is not listed here.

Alternatives recommended

- Use AWS CLI V2 and the `aws` `login` command to use your existing console credentials in the CLI. [Learn more](#)
- Use AWS CloudShell, a browser-based CLI, to run commands. [Learn more](#)

Confirmation

☒ I understand the above recommendation and want to proceed to create an access key.

Step 1

Access key best practices & alternatives

Step 2 - optional

Set description tag

Step 3

Retrieve access keys

Retrieve access keys

If you lose or forget your secret access key, you cannot retrieve it. Instead, create a new access key and make the old key inactive.

Access key

AKIAJ2IW3PKBILMZQR5Z

Secret access key

***** [Show](#)

Access key best practices

- Never store your access key in plain text, in a code repository, or in code.
- Disable or delete access key when no longer needed.
- Enable least-privilege permissions.
- Rotate access keys regularly.

For more details about managing access keys, see the [best practices for managing AWS access keys](#).

[Download .csv file](#)

Done

Identity and Access Management (IAM)

Roles

Policies

IAM users

Root access management

kshitija-admin

Summary

ARN

arn:aws:iam::606175197826:user/kshitija-admin

Access key 1

AKIAJ2IW3PKBDEGX656Z - Active

Never used. Created today.

Created

August 26, 2026, 01:41 (UTC-07:00)

Permissions

Tags

Security credentials

Last Accessed

Access keys (1)

Use access keys to send programmatic calls to AWS from the AWS CLI, AWS Tools for PowerShell, AWS SDKs, or direct AWS API calls. You can have a maximum of two access keys (active or inactive) at a time. [Learn more](#)

AKIAJ2IW3PKBDEGX656Z

Description

Last used

None

Last used region

N/A

Status

Active

Created

2 minutes ago

Last used service

N/A

Actions

API keys (0)

API keys grant access to specific AWS services. You can have a maximum of two API keys per service (active, inactive, or expired) at a time. [Learn more](#)

Service

Any

Status

Any

Actions

Generate API Key

4. Then install AWS CLI

```
Last login: Tue Aug 25 16:47:12 on console
(base) kshitijarohandeshmukh@Kshitijas-MacBook-Air ~ % curl "https://awscli.amazonaws.com/AWSCLIV2.pkg" -o "AWSCLIV2.pkg"
% Total    % Received % Xferd  Average Speed   Time    Time     Current
                                 Dload  Upload   Total   Spent    Left     Speed
100 56.9M  100 56.9M    0     0  93.0M    0 --:--:-- --:--:-- --:--:--  92.9M
(base) kshitijarohandeshmukh@Kshitijas-MacBook-Air ~ % sudo installer -pkg AWSCLIV2.pkg -target /
Password:
installer: Package name is AWS Command Line Interface
installer: Installing at base path /
installer: The install was successful.
(base) kshitijarohandeshmukh@Kshitijas-MacBook-Air ~ % aws --version
aws-cli/2.36.31 Python/3.14.6 Darwin/25.5.0 exe/arm64
(base) kshitijarohandeshmukh@Kshitijas-MacBook-Air ~ %
```

Now let's connect AWS CLI to our account using my Access Key and Secret Key

```
(base) kshitijarohandeshmukh@Kshitijas-MacBook-Air data260-6491 % aws --version
aws-cli/2.36.31 Python/3.14.6 Darwin/25.6.0 exe/arm64
(base) kshitijarohandeshmukh@Kshitijas-MacBook-Air data260-6491 % aws configure
AWS Access Key ID [*****656Z]: 
AWS Secret Access Key [*****5x55]: 
Default region name [us-east-2]:
Default output format [None]:
(base) kshitijarohandeshmukh@Kshitijas-MacBook-Air data260-6491 % aws sts get-caller-identity
{
  "UserId": "AIDAY2IW3PKBJEDQLD6DH",
  "Account": "606175197826",
  "Arn": "arn:aws:iam::606175197826:user/kshitija-admin"
}
(base) kshitijarohandeshmukh@Kshitijas-MacBook-Air data260-6491 %
```

5. Then Install Ollama

```
(base) kshitijarohandeshmukh@Kshitijas-MacBook-Air ~ % ollama --version
ollama version is 0.33.0
(base) kshitijarohandeshmukh@Kshitijas-MacBook-Air ~ % ollama pull qwen3:8b
pulling manifest
pulling a3de06cd1c13: 100% 5.2 GB
verifying sha256 digest
writing manifest
success
(base) kshitijarohandeshmukh@Kshitijas-MacBook-Air ~ % ollama list
NAME      ID              SIZE    MODIFIED
qwen3:8b  500a1f067a9f    5.2 GB   30 seconds ago
(base) kshitijarohandeshmukh@Kshitijas-MacBook-Air ~ % ollama run qwen3:8b
>>> hello
Thinking...
Okay, the user said "hello" and I need to respond. Let me think about how to approach this. First, I should acknowledge their greeting and make them feel welcome. Maybe start with a friendly "Hello!" to be polite. Then, offer assistance since they might have questions or need help. I should keep the tone positive and open-ended to encourage them to ask anything. Also, make sure the response is concise and not too long. Let me check if there's anything else I need to consider. Oh, right, maybe mention that I'm here to help with any questions they have. Yeah, that sounds good. Alright, time to put it all together.
...done thinking.

Hello! 🙌 How can I assist you today? I'm here to help with any questions or tasks you might have!

>>> /bye
(base) kshitijarohandeshmukh@Kshitijas-MacBook-Air ~ %
```


Questions: (Part 1)

1. Create an HTML page titled HW1-{Your Name}

Code:

```

<> index.html X
Homework > HW1 > data260-6491 > code > web_application > index.html > html > head > title
1  <!DOCTYPE html>
2  <html lang="en">
3  <head>
4    <meta charset="UTF-8" />
5    <meta name="viewport" content="width=device-width, initial-scale=1.0" />
6    <title>HW1-Kshitija Shinde</title>
7  <style>
8    body {
9      font-family: Arial, sans-serif;
10     max-width: 600px;
11     margin: 40px auto;
12     padding: 20px;
13     background-color: #f5f5f5;
14     line-height: 1.6;
15   }
16
17   h1 {
18     color: #2e7d32;
19     text-align: center;
20     margin-bottom: 30px;
21     padding-bottom: 10px;
22     border-bottom: 2px solid #2e7d32;
23   }
24
25   form {
26     background: white;
27     padding: 30px;
28     border-radius: 10px;
29     box-shadow: 0 2px 10px rgba(0,0,0,0.1);
30   }
31
32   label {
33     display: block;
34     margin-bottom: 5px;
35     font-weight: bold;
36     color: #333;
37   }
38

```

Output:

Grocery supply and recall notices

Product Name *

Supplier / Brand *

Your Email *

Recall Reason / Description *

Recall Reason Type *

☐ I agree to the terms and conditions.

2. Add a heading tag naming your assigned domain entity (use the header tag that renders the biggest font size).

code:

```
index.html X
Homework > HW1 > data260-6491 > code > web_application > index.html > html > body > h1
70 }
71
72 .checkbox-row label {
73   margin-bottom: 0;
74   font-weight: normal;
75 }
76
77 button {
78   background-color: #2e7d32;
79   color: white;
80   padding: 12px 30px;
81   border: none;
82   border-radius: 5px;
83   font-size: 16px;
84   cursor: pointer;
85   display: block;
86   margin: 20px auto 0;
87 }
88
89 button:hover {
90   background-color: #256428;
91 }
92 </style>
93 </head>
94 <body>
95
96 <!-- this is the main heading for the page, names the domain entity, using h1 since that renders the biggest font size -->
97 <h1>Grocery supply and recall notices</h1>
98
99 <!-- form for reporting a grocery recall, fields match DOMAIN_SCHEMA.md -->
100 <form id="recallForm">
101
```

Output:

The screenshot shows a web form titled "Grocery supply and recall notices" in a green header. The form is a light gray box with rounded corners and a shadow. It contains several input fields with placeholder text: "Product Name *" (placeholder: "e.g. Trader Joe's Organic Frozen Blueberries, 16oz"), "Supplier / Brand *" (placeholder: "who made or supplied it"), "Your Email *" (placeholder: "your email so we can follow up"), and "Recall Reason / Description *" (placeholder: "what happened, when you noticed it, any batch/lot info"). There is a "Recall Reason Type *" dropdown menu with the text "Choose a reason". Below the dropdown is a checkbox labeled "I agree to the terms and conditions." and a green button labeled "Report This Recall". A small gray tooltip with the text "Please fill out this field." points to the "Your Email *" field.

3. Before coding I created DOMAIN_SCHEMA.md so that i can define entity's fields and category values (kind of requirement doc)

code:

```
index.html x DOMAIN_SCHEMA.md x
Homework > HW1 > data260-6491 > DOMAIN_SCHEMA.md > abc # DOMAIN_SCHEMA.md
1 # DOMAIN_SCHEMA.md
2
3 ## Domain: Grocery Supply and Recall Notices
4 ## DOMAIN_ID: 3
5 ## SID4: 6491
6
7 This file defines the fields and category values used in the grocery recall report form built for Part 1 (HTML and JavaScript).
8
9 ---
10
11 ## Entity: Grocery supply and recall notices
12
13 ## Fields
14
15 **1. Product Name** (Primary Field)
16 - Type: required text input
17 - Placeholder: "e.g. Trader Joe's Organic Frozen Blueberries, 16oz"
18 - Behavior: cursor auto-focuses here when the page loads
19 - Example value: `Trader Joe's Organic Frozen Blueberries, 16oz`
20
21 **2. Supplier / Brand** (Secondary Field)
22 - Type: required text input
23 - Placeholder: "who made or supplied it"
24 - Example value: `Trader Joe's (Stevens Creek location)`
25
26 **3. Submitter Email**
27 - Type: required email input
28 - Placeholder: "your email so we can follow up"
29 - Example value: `rohan1@gmail.com`
30
31 **4. Recall Reason / Description** (Content Field)
32 - Type: required textarea
33 - Placeholder: "what happened, when you noticed it, any batch/lot info"
34 - Validation rule: must be more than 25 characters (enforced in JavaScript)
35 - Example value:
36   `grabbed this from the TJ's on Stevens Creek, didn't notice till i got home that the bag had a small tear near the top seal, right where it's supposed to be sealed shut. bunch of the berries at the top were already a little frosted/clumped together like air got in. didn't want to risk it so i didn't open the rest, but the tear looked like it was there before i even picked it up off the shelf`
37
38 **5. Recall Reason Type** (Category Dropdown - 4 options)
39 - Type: select dropdown, required
40 - Options:
41   1. Packaging / Seal Failure
42   2. Contamination
43   3. Undeclared Allergen
44   4. Spoiled or Quality Issue
45
46
47 **6. Terms Agreement**
48 - Type: checkbox, required
49 - Label text: "I agree to the terms and conditions."
50 - Validation rule: must be checked before submission (enforced in JavaScript)
51
52 **7. Submit Button**
53 - Text: "Report This Recall"
54
55 ---
56 ## Notes
57 - All fields marked "required" must have the HTML `required` attribute.
58 - Field 4 (description) and Field 6 (checkbox) are validated using an arrow-function JavaScript validator before allowing submission.
59 - This schema is the blueprint for the HTML form built in Part 1, Question I.4.
```

Output:

The image displays three sequential screenshots of a GitHub repository interface, specifically focusing on the `DOMAIN_SCHEMA.md` file within the `data260-6491` repository. The repository owner is `Kohitja-Shinde9`.

First Screenshot: Shows the top of the file. The domain is **Domain: Grocery Supply and Recall Notices** and the ID is **DOMAIN_ID: 3**. The file is titled **SID4: 6491**. A description states: "This file defines the fields and category values used in the grocery recall report form built for Part 1 (HTML and JavaScript)." The entity is **Entity: Grocery supply and recall notices**. The first field is **1. Product Name (Primary Field)**, which is a required text input with a placeholder "e.g. Trader Joe's Organic Frozen Blueberries, 16oz", cursor auto-focus, and an example value "Trader Joe's Organic Frozen Blueberries, 16oz". The second field is **2. Supplier / Brand (Secondary Field)**.

Second Screenshot: Continues from the first field. The second field **2. Supplier / Brand (Secondary Field)** is a required text input with a placeholder "who made or supplied it" and an example value "Trader Joe's (Stevens Creek location)". The third field is **3. Submitter Email**, a required email input with a placeholder "your email so we can follow up" and an example value "rohan@gmail.com". The fourth field is **4. Recall Reason / Description (Content Field)**, a required textarea with a placeholder "what happened, when you noticed it, any batch/lot info", a validation rule "must be more than 25 characters (enforced in JavaScript)", and an example value describing a recall of frozen blueberries. The fifth field is **5. Recall Reason Type (Category Dropdown - 4 options)**, a required select dropdown with options: I. Packaging / Seal Failure, II. Contamination, III. Undeclared Allergen, and IV. Spoiled or Quality Issue. The sixth field is **6. Terms Agreement**, a required checkbox with a label "I agree to the terms and conditions." and a validation rule "must be checked before submission (enforced in JavaScript)".

Third Screenshot: Continues from the sixth field. The seventh field is **7. Submit Button**, a text button with the text "Report This Recall". The **Notes** section at the bottom states: "All fields marked 'required' must have the HTML `required` attribute. Field 4 (description) and Field 6 (checkbox) are validated using an arrow-function JavaScript validator before allowing submission. This schema is the blueprint for the HTML form built in Part 1, Question 1.4."

4. Create a form for your domain entity. The form should include the following:
- a) A required text input for the entity's primary field, with a placeholder describing it. The cursor should automatically focus on this field when the page loads.
 - b) A required text input for the entity's secondary field, with a descriptive placeholder.
 - c) A required email input for the submitter's email, with a descriptive placeholder.
 - d) A required text area for the entity's content/description field, with a descriptive placeholder.
 - e) A dropdown for category selection, with four domain-appropriate options.
 - f) A checkbox and a label with the text "I agree to the terms and conditions."
 - g) A submit button with domain-appropriate text

Code

```
index.html x DOMAIN_SCHEMA.md x
Homework > HW1 > data260-6491 > code > web_application > index.html > html > body > form#recallForm
95
96 <!-- this is the main heading for the page, names the domain entity, using h1 since that renders the biggest font size -->
97 <h1>Grocery supply and recall notices</h1>
98
99 <!-- form for reporting a grocery recall, fields match DOMAIN_SCHEMA.md -->
100 <form id="recallForm">
101
102 <!-- primary field, product name, required, autofocus so cursor lands here on page load -->
103 <label for="productName">Product Name *</label>
104 <input type="text" id="productName" placeholder="e.g. Trader Joe's Organic Frozen Blueberries, 16oz" required autofocus />
105
106 <!-- secondary field, supplier or brand, required -->
107 <label for="supplierName">Supplier / Brand *</label>
108 <input type="text" id="supplierName" placeholder="who made or supplied it" required />
109
110 <!-- submitter email, required -->
111 <label for="submitterEmail">Your Email *</label>
112 <input type="email" id="submitterEmail" placeholder="your email so we can follow up" required />
113
114 <!-- content or description field, required textarea, needs more than 25 characters checked in js -->
115 <label for="recallDescription">Recall Reason / Description *</label>
116 <textarea id="recallDescription" placeholder="what happened, when you noticed it, any batch/lot info" required></textarea>
117
118 <!-- category dropdown with 4 domain appropriate options -->
119 <label for="recallType">Recall Reason Type *</label>
120 <select id="recallType" required>
121   <option value="" disabled selected>Choose a reason</option>
122   <option value="Packaging / Seal Failure">Packaging / Seal Failure</option>
123   <option value="Contamination">Contamination</option>
124   <option value="Undeclared Allergen">Undeclared Allergen</option>
125   <option value="Spoiled or Quality Issue">Spoiled or Quality Issue</option>
126 </select>
127
128 <!-- checkbox and label for terms and conditions, checked in js before allowing submit -->
129 <div class="checkbox-row">
130   <input type="checkbox" id="agreeTerms" />
131   <label for="agreeTerms">I agree to the terms and conditions.</label>
132 </div>
133
134 <!-- submit button with text that fits the grocery recall domain -->
135 <button type="submit">Report This Recall</button>
136
137 </form>
138
139 <!-- linking the javascript file for part 2, placed at the end of the html file as required -->
140 <script src="script.js"></script>
141 </body>
142 </html>
```

Output

Grocery supply and recall notices

Product Name *
e.g. Trader Joe's Organic Frozen Blueberries, 16oz

Supplier / Brand *
who made or supplied it

Your Email *
your email so we can follow up

Recall Reason / Description *
what happened, when you noticed it, any batch/lot info

Recall Reason Type *
Choose a reason

☐ I agree to the terms and conditions.

Report This Recall

Grocery supply and recall notices

Product Name *
e.g. Trader Joe's Organic Frozen Blueberries, 16oz

Supplier / Brand *
who made or supplied it

Your Email *
your email so we can follow up

Recall Reason / Description *
what happened, when you noticed it, any batch/lot info

Recall Reason Type *
Choose a reason

- ✓ Choose a reason
- Packaging / Seal Failure
- Contamination
- Undeclared Allergen
- Spoiled or Quality Issue

5. Add a script tag to link your JavaScript code for Part II at the end of your HTML file.

```
139 <!-- linking the javascript file for part 2, placed at the end of the html file as required -->
140 <script src="script.js"></script>
141 </body>
142 </html>
```

II. Javascript

1. Write a JavaScript function using an arrow function to validate:
 - a) Verify that the content field has more than 25 characters. If validation fails, display an alert with an appropriate message.
 - b) Verify that the terms and conditions checkbox is checked. If validation fails, display an alert with an appropriate message.
2. After the form submission is successful, convert the form data into a JSON string and log the output in the console.
3. Use object destructuring to extract the primary field and email field from the parsed object and log their values in the console.
4. Use the spread operator to add a new field submissionDate with the current date and time to the parsed object. Log the updated parsed object in the console.
5. Create a closure to track how many times the form has been successfully submitted and log the submission count each time the form is submitted.

Code:

```
< index.html x DOMAIN_SCHEMA.md x JS script.js x
Homework > HW1 > data260-6491 > code > web_application > JS script.js > submissionCounter > <function>
1 // keeping a running count of submissions here, this stays private and only the counter can touch it
2 const submissionCounter = (() => {
3     let count = 0;
4     return () => {
5         count = count + 1;
6         return count;
7     };
8 })();
9
10 // two checks in one place, description length and the checkbox. throwing an alert if either one is off
11 const validateForm = (descriptionValue, isChecked) => {
12     const trimmedDescription = descriptionValue.trim();
13
14     if (trimmedDescription.length <= 25) {
15         alert("please write a bit more in the recall reason, needs to be more than 25 characters");
16         return false;
17     }
18
19     if (!isChecked) {
20         alert("you need to check the box agreeing to the terms and conditions before submitting");
21         return false;
22     }
23
24     return true;
25 };
26
27 document.getElementById("recallForm").addEventListener("submit", (event) => {
28     event.preventDefault();
29
30     // trimming everything on the way in so stray spaces don't sneak through as real input
31     const productName = document.getElementById("productName").value.trim();
32     const supplierName = document.getElementById("supplierName").value.trim();
33     const submitterEmail = document.getElementById("submitterEmail").value.trim();
34     const recallDescription = document.getElementById("recallDescription").value.trim();
35     const recallType = document.getElementById("recallType").value;
36     const agreeTerms = document.getElementById("agreeTerms").checked;
37
38     // if this comes back false just stop right here, nothing below should run
39     const isValid = validateForm(recallDescription, agreeTerms);
40     if (!isValid) {
41         return;
42     }
43
44     const formData = {
45         productName: productName,
46         supplierName: supplierName,
47         submitterEmail: submitterEmail,
48         recallDescription: recallDescription,
49         recallType: recallType
```

<> index.html X DOMAIN_SCHEMA.md X JS script.js X

Homework > HW1 > data260-6491 > code > web_application > JS script.js > submissionCounter > <function>

```
27 document.getElementById("recallForm").addEventListener("submit", (event) => {
28     event.preventDefault();
29
30     // trimming everything on the way in so stray spaces don't sneak through as real input
31     const productName = document.getElementById("productName").value.trim();
32     const supplierName = document.getElementById("supplierName").value.trim();
33     const submitterEmail = document.getElementById("submitterEmail").value.trim();
34     const recallDescription = document.getElementById("recallDescription").value.trim();
35     const recallType = document.getElementById("recallType").value;
36     const agreeTerms = document.getElementById("agreeTerms").checked;
37
38     // if this comes back false just stop right here, nothing below should run
39     const isValid = validateForm(recallDescription, agreeTerms);
40     if (!isValid) {
41         return;
42     }
43
44     const formData = {
45         productName: productName,
46         supplierName: supplierName,
47         submitterEmail: submitterEmail,
48         recallDescription: recallDescription,
49         recallType: recallType
50     };
51
52     const jsonString = JSON.stringify(formData);
53     console.log("form data as JSON string:", jsonString);
54
55     const parsedData = JSON.parse(jsonString);
56
57     // just pulling out product name and email here, renaming them so it reads a bit clearer
58     const { productName: reportedProduct, submitterEmail: reportedEmail } = parsedData;
59     console.log("product name:", reportedProduct);
60     console.log("submitter email:", reportedEmail);
61
62     // copying everything over and tacking on when this actually got submitted
63     const updatedData = { ...parsedData, submissionDate: new Date().toString() };
64     console.log("updated data with submission date:", updatedData);
65
66     const currentCount = submissionCounter();
67     console.log("this form has been submitted", currentCount, "time(s) so far");
68
69     alert("thanks, your recall report was submitted");
70
71     // clearing everything out so the form is ready to go again
72     document.getElementById("recallForm").reset();
73 });
```

Output:

Grocery supply and recall notices

Product Name *

Supplier / Brand *

Your Email *

Recall Description *

 Please enter a part following '@'. 'example@' is incomplete.

what happened, when you noticed it, any batch/lot info

Recall Reason Type *

Choose a reason

☐ I agree to the terms and conditions.

Report This Recall

Grocery supply and recall notices

Product Name *

Supplier / Brand *

Your Email *

Recall Reason / Description *

what happened, when you noticed it, any batch/lot info

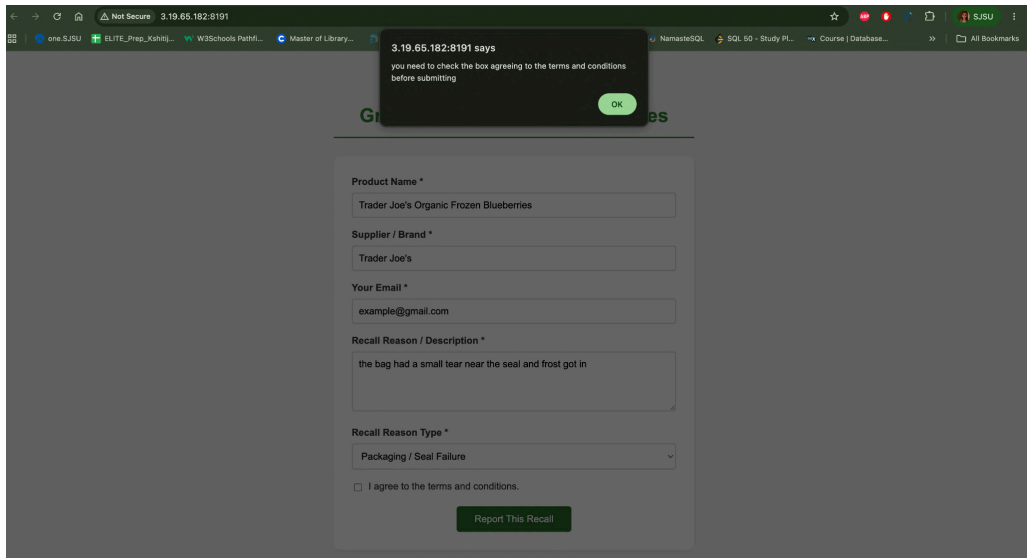
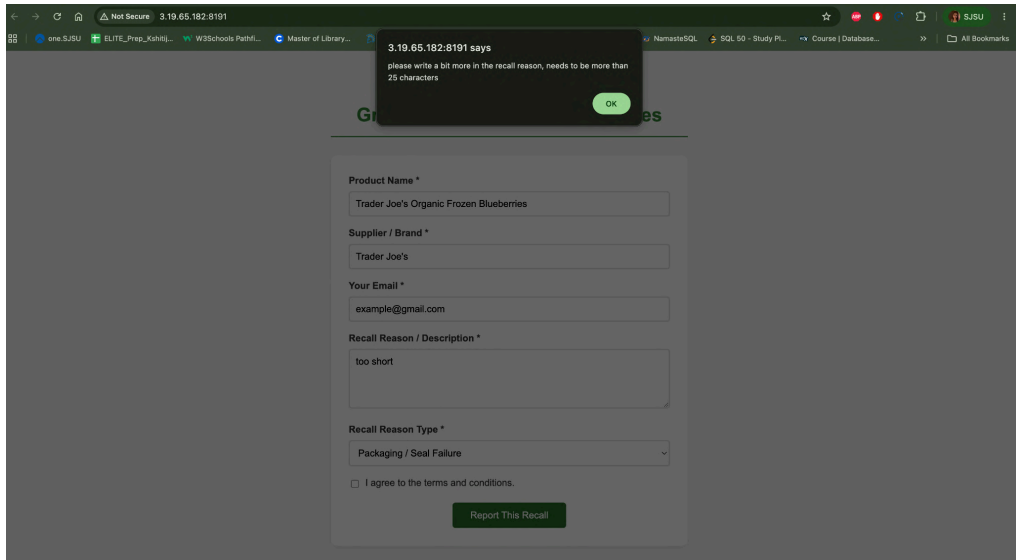
Recall Reason Type *

 Please fill out this field.

Packaging / Seal Failure

☐ I agree to the terms and conditions.

Report This Recall



Successfully submitted

3.19.65.182:8191 says
thanks, your recall report was submitted

Grocery supply and recall notices

Product Name *
Trader Joe's Organic Frozen Blueberries

Supplier / Brand *
Trader Joe's

Your Email *
example@gmail.com

Recall Reason / Description *
the bag had a small tear near the seal and frost got in

Recall Reason Type *
Packaging / Seal Failure

☒ I agree to the terms and conditions.

Report This Recall

Grocery supply and recall notices

Product Name *
e.g. Trader Joe's Organic Frozen Blueberries, 16oz

Supplier / Brand *
who made or supplied it

Your Email *
your email so we can follow up

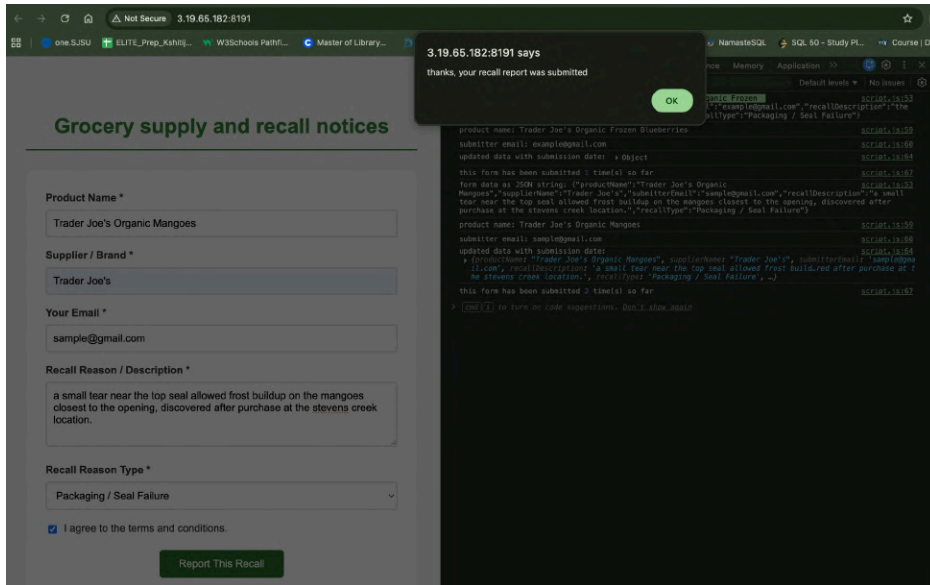
Recall Reason / Description *
what happened, when you noticed it, any batch/lot info

Recall Reason Type *
Choose a reason

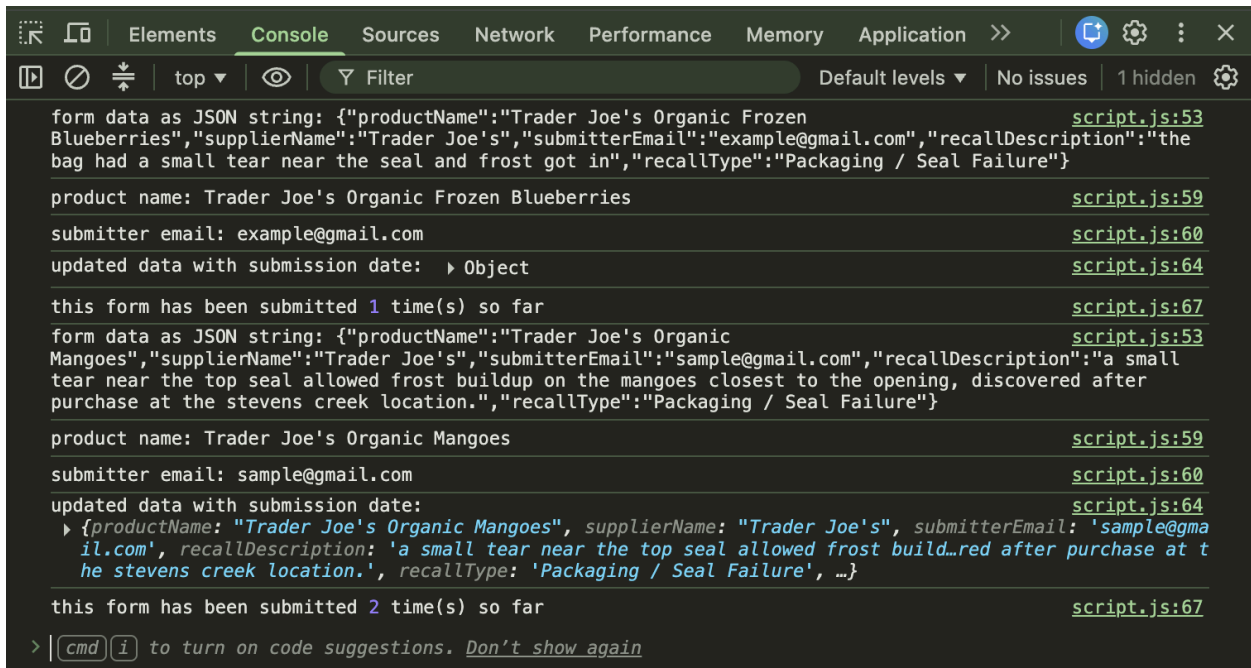
☐ I agree to the terms and conditions.

Report This Recall

```
fer data as JSON string: {"productName":"Trader Joe's Organic Frozen Blueberries","supplierName":"Trader Joe's","submitterEmail":"example@gmail.com","recallDescription":"the bag had a small tear near the seal and frost got in","recallType":"Packaging / Seal Failure"} script:15:53
product name: Trader Joe's Organic Frozen blueberries script:15:58
submitter email: example@gmail.com script:15:68
updated date with submission date: 1 Object script:15:64
this form has been submitted 1 time(s) so far script:15:67
> [cmd] to turn on code suggestions. Don't show again
```

Created a closure to track how many times the form has been successfully submitted and log the submission count each time the form is submitted.



Docker: Create a docker image of the above application and build and run the application using docker in your local.

```
(base) kshiti.jarohandeshmukh@KshitiHijas-MacBook-Air data260-6491 % docker build -t grocery-recall-app .
[+] Building 6.7s (9/9) FINISHED
=> [internal] load build definition from Dockerfile                                0.0s
=> => transferring dockerfile: 344B                                              0.0s
=> [internal] load metadata for docker.io/library/nginx:alpine                  4.9s
=> [auth] library/nginx:pull token for registry-1.docker.io                     0.0s
=> [internal] load .dockerignore                                                 0.0s
=> => transferring context: 2B                                                  0.0s
=> [1/3] FROM docker.io/library/nginx:alpine:sha256:db35bfc6b2951ef7f8a72db5db120288c1277faeeb4a6d4b95a26fead017d5913 1.4s
=> => resolve docker.io/library/nginx:alpine:sha256:db35bfc6b2951ef7f8a72db5db120288c1277faeeb4a6d4b95a26fead017d5913 0.0s
=> sha256:6ecb77a8803aca4c7f9a8e821d47479b7fae5f1e448123dae88b032bea88b34 404B / 404B 0.4s
=> sha256:c3ee22b57f6bdfbf1e048d52d0822a667b855270efd4843ab57bb8b62bf00967 20.07MB / 20.07MB 0.5s
=> sha256:aca9e7c5ccc193046bfa55546d17a58741d5fa7b6354f9db9998e9ec4871767f4 1.40kB / 1.40kB 0.4s
=> sha256:7feeb37758e4f084364db6a9ea73d0b8525e4de269809af82f2be15491dbfff 1.21kB / 1.21kB 0.4s
=> sha256:ed0b77a8803aca4c7f9a8e821d47479b7fae5f1e448123dae88b032bea88b34 955B / 955B 0.3s
=> sha256:3be296cd3c979a33094d1fabe554c088a2b2c9957614f1b03a7a169ce9d062 628B / 628B 0.0s
=> sha256:1b000889ffff63564828cd0644838439818b178439d0ea8f9ef3805429540170 1.92MB / 1.92MB 0.0s
=> sha256:5de55e5ef9c03997441461efe7ba23a986db059c0bb78b38f84ee0d72b99167 4.18MB / 4.18MB 0.5s
=> extracting sha256:5de55e5ef9c03997441461efe7ba23a986db059c0bb78b38f84ee0d72b99167 0.1s
=> extracting sha256:1b000889ffff63564828cd0644838439818b178439d0ea8f9ef3805429540170 0.1s
=> extracting sha256:3be296cd3c979a33094d1fabe554c088a2b2c9957614f1b03a7a169ce9d062 0.0s
=> extracting sha256:ed0b77a8803aca4c7f9a8e821d47479b7fae5f1e448123dae88b032bea88b34 0.0s
=> extracting sha256:7feeb37758e4f084364db6a9ea73d0b8525e4de269809af82f2be15491dbfff 0.0s
=> extracting sha256:aca9e7c5ccc193046bfa55546d17a58741d5fa7b6354f9db9998e9ec4871767f4 0.0s
=> extracting sha256:c3ee22b57f6bdfbf1e048d52d0822a667b855270efd4843ab57bb8b62bf00967 0.3s
=> [internal] load build context                                                0.0s
=> => transferring context: 44.09kB                                             0.0s
=> [2/3] RUN rm -rf /usr/share/nginx/html/*                                   0.2s
=> [3/3] COPY ./usr/share/nginx/html/                                         0.1s
=> exporting to image                                                         0.0s
=> => exporting layers                                                         0.0s
=> => exporting manifest sha256:e2d75b1cb42dd4c6686719bb109312056dc17d1f32e511739d1aa3ebe95f31a9 0.0s
=> => exporting config sha256:38748d7728b7dfeea9ff1147b954a078254eacc4c764cb1403facb08f01de725 0.0s
=> => exporting attestation manifest sha256:33c3381f4a94cac52ba274ea0218406bee8b207ede5419e0b586d43280795d8 0.0s
=> => exporting manifest list sha256:f7a149d2159e56c57fc125d814894a97922b42b31715985dded5170362f134de 0.0s
=> => naming to docker.io/library/grocery-recall-app:latest                  0.0s
=> => unpacking to docker.io/library/grocery-recall-app:latest              0.0s

View build details: docker-desktop://dashboard/build/desktop-linux/desktop-linux/uhyymmecptd5wh76t2yvvzst38f
(base) kshiti.jarohandeshmukh@KshitiHijas-MacBook-Air data260-6491 % docker run -d -p 8191:80 --name grocery-recall-container grocery-recall-app
05f8748d5b140dc9b7708b678cf21be74a4fd34b7c8d5836d8261e3a4839b3
(base) kshiti.jarohandeshmukh@KshitiHijas-MacBook-Air data260-6491 % docker ps
CONTAINER ID   IMAGE                                COMMAND                  CREATED          STATUS          PORTS          NAMES
05f8748d5b14   grocery-recall-app                  "/docker-entrypoint..." 2 minutes ago   Up 2 minutes   0.0.0.0:8191->80/tcp, [::]:8191->80/tcp   grocery-recall-container
(base) kshiti.jarohandeshmukh@KshitiHijas-MacBook-Air data260-6491 %
```

```
(base) kahlitj@rohandshehuahKahlitijas-MacBook-Air: data260-6491 % aws ec2 get-login-password --region us-east-2 | docker login --username AWS --password-stdin 606175197826.dkr.ecr.us-east-2.amazonaws.com
Login Succeeded
(base) kahlitj@rohandshehuahKahlitijas-MacBook-Air: data260-6491 % docker buildx build --platform linux/amd64 -t 606175197826.dkr.ecr.us-east-2.amazonaws.com/grocery-recall-app:latest --push .
[+] Building 3.5s (11/21) FINISHED
x
docker:desktop-linu
» [internal] load build definition from Dockerfile
s
» » transferring dockerfile: 522B
s
» [internal] load metadata for docker.io/library/nginx:alpine
s
» [auth] library/nginx:pull token for registry-1.docker.io
s
» [internal] load .dockerignore
s
» » transferring context: 2B
s
» [1/4] FROM docker.io/library/nginx:alpine@sha256:db35bfc6b2951e7f8a72db5db120288c127ffae6b46d4b95a26fead017d5913
s
» » resolve docker.io/library/nginx:alpine@sha256:db35bfc6b2951e7f8a72db5db120288c127ffae6b46d4b95a26fead017d5913
s
» [internal] load build context
s
» » transferring context: 5.92kB
s
» CACHED [2/4] RUN rm -rf /usr/share/nginx/html/*
s
» [3/4] COPY /usr/share/nginx/html/
s
» [4/4] RUN sed -i 's/listen.*00;/listen 8191;/' /etc/nginx/conf.d/default.conf
s
» exporting to image
s
» » exporting layers
s
» » exporting manifest sha256:4e23c0e5ba3bad80eb72b5813dd7bd571f88ddc2d059774d74e694c8476ce18
s
» » exporting config sha256:49ab56ffad3c642cc91509c7c1ab2e3f9ebad1d99ba16a2d91a3d28c40d2f39
s
» » exporting attestation manifest sha256:71ae7695b69e4db914919a9e2835d69a343b3793acdc3774bcf9c2481b1e34f
s
» » exporting manifest list sha256:1a5824ada8038ae48795bdc2458ef9cb93c357afdd5420ae088c95826b1d49
s
» » naming to 606175197826.dkr.ecr.us-east-2.amazonaws.com/grocery-recall-app:latest
s
» » pushing layers
s
» » pushing manifest for 606175197826.dkr.ecr.us-east-2.amazonaws.com/grocery-recall-app:latest@sha256:1a5824ada8038ae48795bdc2458ef9cb93c357afdd5420ae088c95826b1d49
s
» [auth] sharing credentials for 606175197826.dkr.ecr.us-east-2.amazonaws.com
s
View build details: docker-desktop://dashboard/build/desktop-linux/desktop-linux/zosj32nkevjkxak2n1zmjkg9o
(base) kahlitj@rohandshehuahKahlitijas-MacBook-Air: data260-6491 % docker ps
CONTAINER ID        IMAGE               COMMAND                  CREATED            STATUS            PORTS                    NAMES
00787a8d8b14        grocery-recall-app  "/docker-entrypoint. ..."  4 hours ago       Up 4 hours        0.0.0.0:8191->800/tcp, [::]:8191->800/tcp  grocery-recall-container
```

docker.desktopPERSONAL

Search

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K

Ask GordonBETA

Containers

Images

Volumes

Kubernetes

Builds

Models

MCP ToolkitBETA

Docker Hub

Docker Scout

Extensions

ContainersGive feedback

Container CPU usage0.00% / 1000% (10 CPUs available)

Container memory usage8.91MB / 7.47GB

Show charts

Search

Only show running containers

	Name	Container ID	Image	Port(s)	CPU (%)	Memory usage...	Memory (%)	Disk read/w	Actions
☐	grocery-recall-container	05f8748d5b14	grocery-recall-app	8191.80	0%	8.91MB / 7.65GB	0.11%	0B / 12.3KB	
☐	fuel_price_project	-	-	-	0%	0B / 0B	0%	0B / 0B	

Showing 2 items

Engine runningRAM 1.13 GB CPU 0.30% Disk: 5.38 GB used (limit 223.63 GB)

TerminalOutdated version

Docker image

docker.desktopPERSONAL

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LocalMy Hub

2.13 GB / 4.01 GB in use 5 images

Last refresh: 4 hours ago

Search

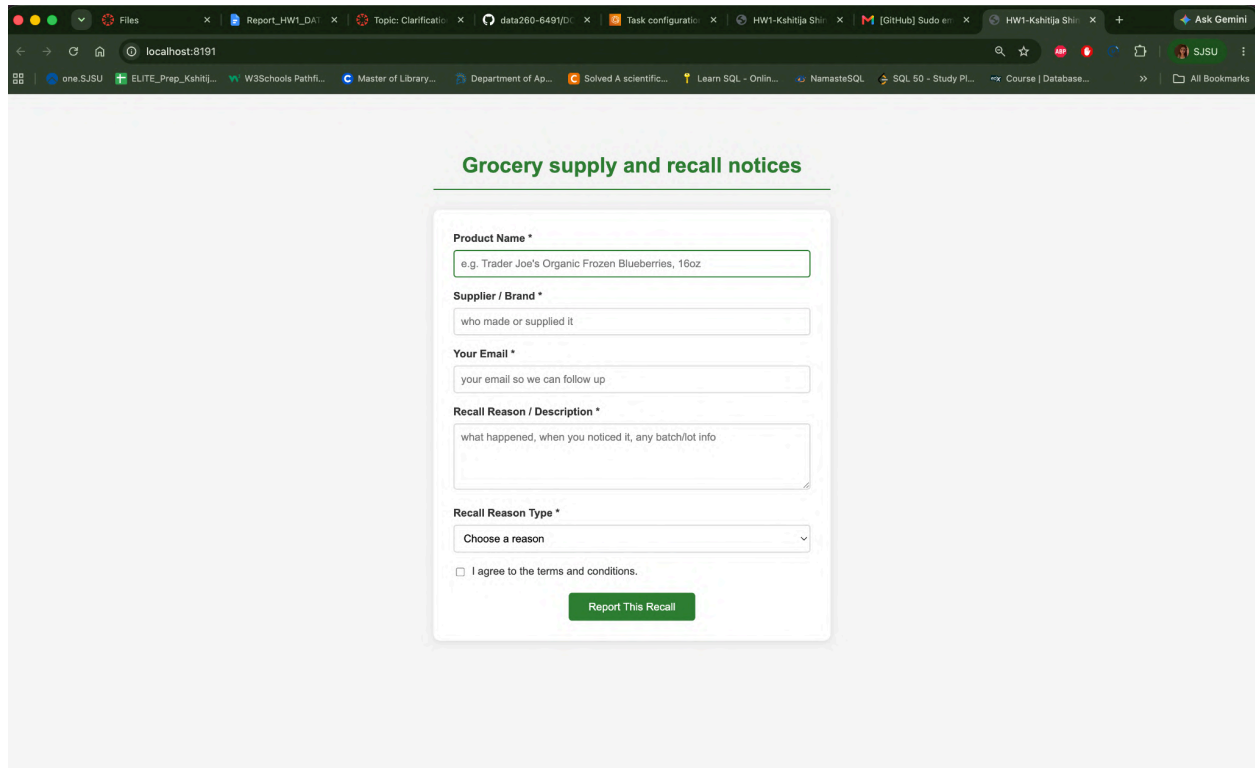
	Name	Tag	Image ID	Created	Size	Actions
☐	postgres	13	4689940c6838	10 months ago	645.43 MB	
☐	apache/airflow	2.10.1	41928f36b625	2 years ago	1.97 GB	
☐	ghcr.io/mlflow/mlflow	v2.13.0	47c5c26d06a4	2 years ago	1.07 GB	
☐	grocery-recall-app	latest	f7a149d2159e	4 hours ago	93.01 MB	
☐	606175197826.dkr.ecr.us-east-2.amazonaws.com	latest	1a582a4ada08	25 minutes ago	26.29 MB	

Showing 5 items

Engine runningRAM 1.13 GB CPU 0.10% Disk: 5.38 GB used (limit 223.63 GB)

TerminalOutdated version

Provide the screenshot of your app running on your localhost.



AWS ECS: Create an AWS ECS service (only one task) running the above application using the docker image created above.

1. Creating private repository

The screenshot shows the 'Create private repository' page in the AWS Management Console. The page is divided into three main sections: General settings, Image tag settings, and Encryption settings. The General settings section includes a 'Repository name' field with the value '606175197826.dkr.ecr.us-east-2.amazonaws.com/grocery-recall-app'. The Image tag settings section includes 'Image tag mutability' (set to 'Mutable') and 'Mutable tag exclusions'. The Encryption settings section includes a warning message and 'Encryption configuration' (set to 'AES-256'). The right sidebar contains a 'Workflows' section and a 'Why use workflows' section. The footer includes navigation links and copyright information.

General settings

Repository name
Enter a concise name. Repositories support namespaces, which you can use to group similar repositories.
606175197826.dkr.ecr.us-east-2.amazonaws.com/grocery-recall-app
18 out of 256 characters maximum (2 minimum). The name must start with a letter and can only contain lowercase letters, numbers, and special characters _./.

Image tag settings [Info](#)

Image tag mutability
Choose the tag mutability setting.
☒ **Mutable**
Image tags can be overwritten.
☐ **Immutable**
Image tags can't be overwritten.

Mutable tag exclusions
Tags that match these filters will be immutable (can't be overwritten). Using wildcards (*) will match zero or more image tag characters.

Filters must only contain letters, numbers, and special characters (.,*). Each filter is limited to 128 characters, 2 wildcards (*), and you can add up to 5 filters in the exclusion list.
[Add filter](#)

Encryption settings [Info](#)

Encryption configuration
By default, repositories use the industry standard Advanced Encryption Standard (AES) encryption. You can optionally choose to use a key stored in the AWS Key Management Service (KMS) to encrypt the images in your repository.
☒ **AES-256**
Industry standard Advanced Encryption Standard (AES) encryption
☐ **AWS KMS**
AWS Key Management Service (KMS)

Workflows

Workflows walk you through common tasks with automatic navigation and pre-filled values. You can customize these pre-filled defaults as you go.

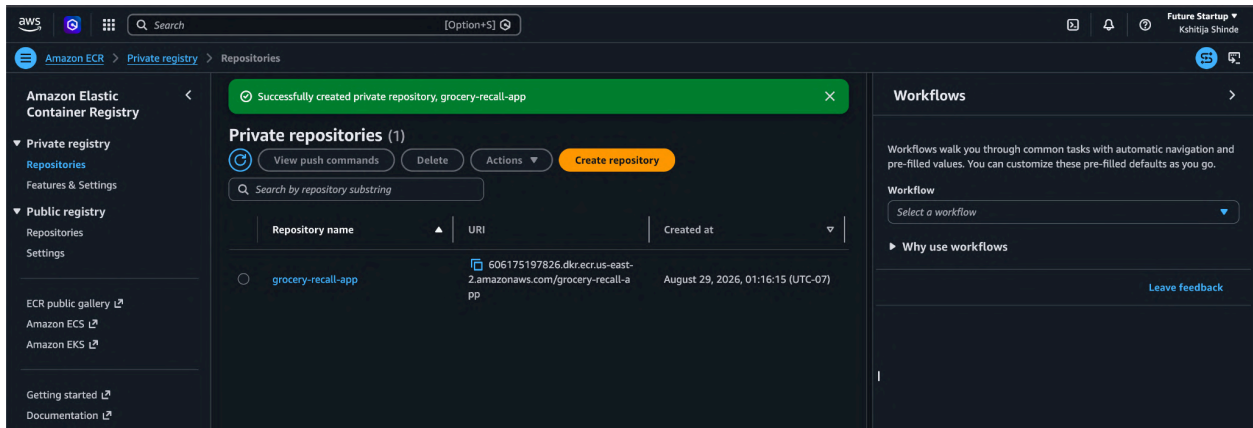
Workflow
Select a workflow

Why use workflows

[Leave feedback](#)

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ECR created



Images pushed to ECR



Created security group

aws

Search

[Option+S]

EC2 > Security Groups > Create security group

0

Future Startup
Kshitija Shinde

Create security group Info

A security group acts as a virtual firewall for your instance to control inbound and outbound traffic. To create a new security group, complete the fields below.

Basic details

Security group name Info

grocery-recall-sg

Name cannot be edited after creation.

Description Info

allows web traffic for grocery recall app

VPC Info

vpc-0b2df13c4b3c6fbc9

Inbound rules Info

Type Info

Protocol Info

Port range Info

Source Info

Description - optional Info

Custom TCP

TCP

8191

A...

0.0.0.0/0

Delete

Add rule

Outbound rules Info

Type Info

Protocol Info

Port range Info

Destination Info

Description - optional Info

All traffic

All

All

C...

0.0.0.0/0

Delete

Add rule

Workflows

Workflows walk you through common tasks with automatic navigation and pre-filled values. You can customize these pre-filled defaults as you go.

Workflow

Select a workflow

Why use workflows

Leave feedback

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aws

Search

[Option+S]

EC2 > Security Groups > sg-09b7add9de89d358c

0

Future Startup
Kshitija Shinde

EC2

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AMI Catalog

Elastic Block Store

Volumes

Snapshots

Lifecycle Manager

Network & Security

Security Groups

Elastic IPs

Placement Groups

Key Pairs

Network Interfaces

Security group (sg-09b7add9de89d358c | grocery-recall-sg) was created successfully

Details

sg-09b7add9de89d358c - grocery-recall-sg

Actions

Details

Security group name

grocery-recall-sg

Security group ID

sg-09b7add9de89d358c

Description

allows web traffic for grocery recall app

VPC ID

vpc-0b2df13c4b3c6fbc9

Owner

606175197826

Inbound rules count

1 Permission entry

Outbound rules count

1 Permission entry

Inbound rules

Outbound rules

Sharing

VPC associations

Related resources - new

Tags

Inbound rules (1)

Manage tags

Edit inbound rules

Search

< 1 >

⚙

☐

Name

Security group rule ID

IP version

Type

Protocol

☐

-

sg-0d9daf7b4376769c8

IPv4

Custom TCP

TCP

Workflows

Workflows walk you through common tasks with automatic navigation and pre-filled values. You can customize these pre-filled defaults as you go.

Workflow

Select a workflow

Why use workflows

Leave feedback

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Now create a ECS Cluster

The screenshot shows the 'Create cluster' page in the AWS Management Console. The left sidebar contains navigation links for 'Amazon Elastic Container Service', 'Favorites and recents', 'Express Mode', 'Clusters', 'Namespaces', 'Task definitions', 'Daemon task definitions', 'Account settings', 'AWS Resource Explorer', 'AWS Batch', 'Amazon ECR', 'Repositories', 'Online learning workshop', 'Documentation', 'Discover products', 'Subscriptions', and 'Tell us what you think'. The main content area is titled 'Create cluster' and includes an 'Info' icon. It explains that an Amazon ECS cluster groups tasks, services, and allows for shared capacity. Below this, the 'Cluster configuration' section has a 'Cluster name' input field with the value 'grocery-recall-cluster' and a note that the name must be 1 to 255 characters. A 'Service Connect defaults - optional' link is also present. The 'Infrastructure - advanced' section offers three options for obtaining compute capacity: 'Fargate only' (selected), 'Fargate and Managed Instances', and 'Fargate and Self-managed instances'. Below these are sections for 'Monitoring - optional' and 'Encryption - optional'. The right sidebar shows 'Workflows' and a 'Why use workflows' section. The footer includes 'CloudShell', 'Agent Toolkit for AWS', 'Feedback', 'Language', and copyright information for 2026 Amazon Web Services, Inc.

The screenshot shows the 'Clusters' page in the AWS Management Console. A green success message at the top states 'Cluster grocery-recall-cluster has been created successfully.' with 'View cluster' and 'Create cluster' buttons. The 'Clusters (1)' section includes a search bar and a table of clusters. The table has columns for 'Cluster', 'Services', 'Tasks', and 'Container instances'. The 'grocery-recall-cluster' is listed with 0 services, 'No tasks running', and 0 EC2 instances. The right sidebar shows 'Workflows' and a 'Why use workflows' section. The footer is identical to the previous screenshot.

Cluster	Services	Tasks	Container instances
grocery-recall-cluster	0	No tasks running	0 EC2

Amazon Elastic Container Service > Clusters > grocery-recall-cluster > Services

Cluster grocery-recall-cluster has been created successfully. [View cluster](#)

grocery-recall-cluster

Last updated August 29, 2026, 01:43 (UTC-7:00)

Cluster overview

Cluster ARN arn:aws:ecs:us-east-1:123456789012:cluster/grocery-recall-cluster	Status Active	CloudWatch monitoring Default	Action logs Turned off
Services 0 active 0 draining	Tasks 0 running 0 pending	Registered container instances -	

Services (0)

Last updated August 29, 2026, 01:43 (UTC-7:00)

Manage tags Update Delete service Create

Filter services by value Filter launch type Any launch type Filter scheduling strategy Any scheduling strategy Filter resource management type Any resource management type

Service name	ARN	Status	Schedu...	L...
No services				
No services to display.				

Create

Workflows

Workflows walk you through common tasks with automatic navigation and pre-filled values. You can customize these pre-filled defaults as you go.

Workflow Select a workflow

Why use workflows

Leave feedback

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Now task definition

Amazon Elastic Container Service > Task definitions > grocery-recall-task > Revision 2 > Containers

Task definition successfully created
grocery-recall-task-2 has been successfully created. You can use this task definition to deploy a service or run a task. [View task definition](#)

grocery-recall-task:2

Last updated August 29, 2026, 03:34 (UTC-7:00)

Overview

ARN arn:aws:ecs:us-east-2:606175197826:task-definition/grocery-recall-task-2	Status ACTIVE	Time created August 29, 2026, 03:34 (UTC-7:00)	App environment Fargate
Task role -	Task execution role ecsTaskExecutionRole	Operating system/Architecture Linux/X86_64	Network mode awsvpc
Fault injection Turned off			

Containers JSON Task placement Volumes (0) Requires attributes Tags

Task size

Task CPU 256 units (0.25 vCPU)

Task CPU maximum allocation for containers

CPU (unit) 0 20 40 60 80 100 120 140 160 180 200 220 240

grocery-recall-container Shared task CPU

Task memory 512 MiB (0.5 GiB)

Task memory maximum allocation for container memory reservation

Memory (MiB) 0 50 100 150 200 250 300 350 400 450 500

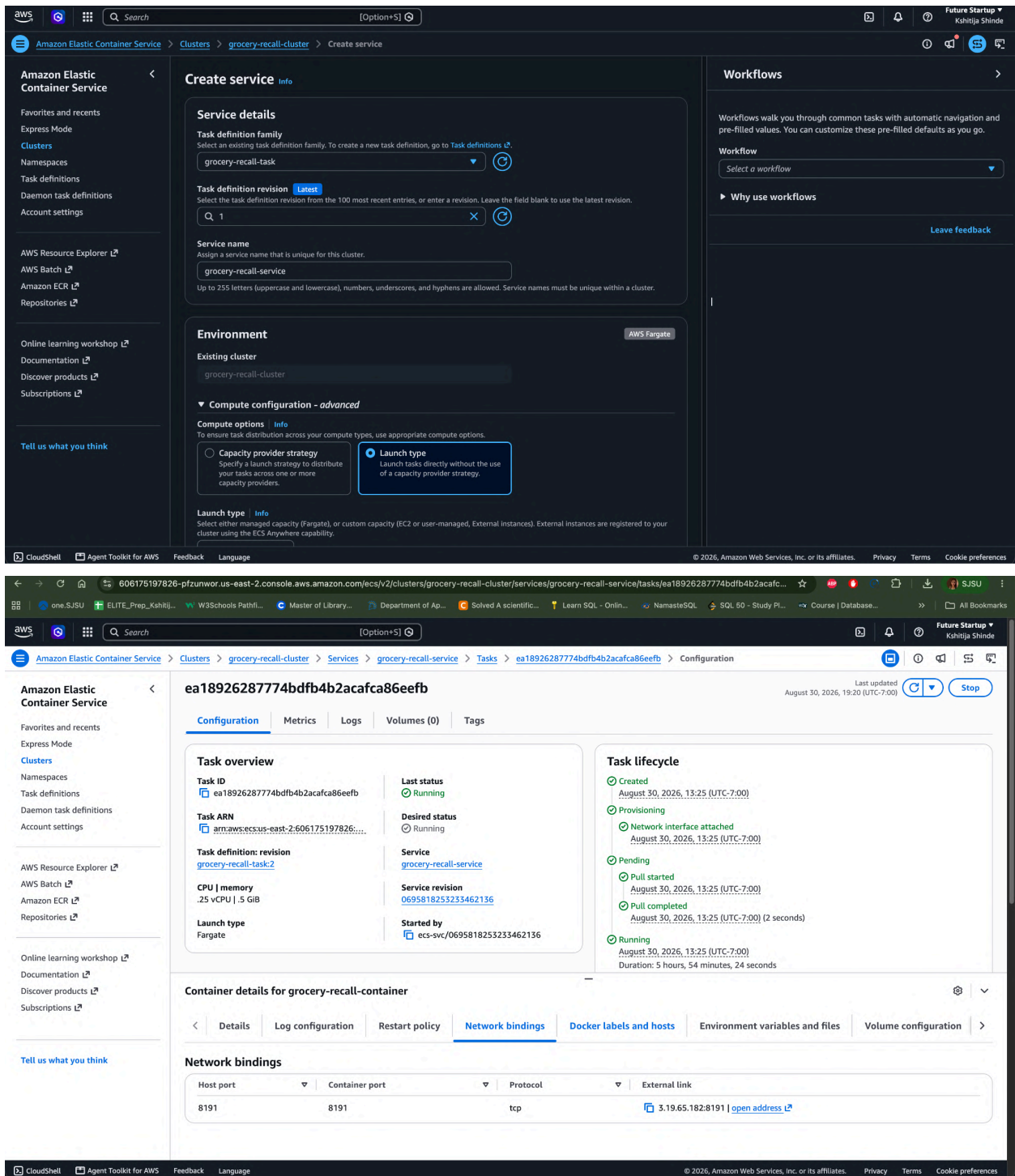
grocery-recall-container Shared task memory

Container: grocery-recall-container

Essential container

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Created ECS service



The screenshot displays the AWS Management Console interface for creating a new ECS service. The left sidebar shows the navigation menu with options like 'Amazon Elastic Container Service', 'Favorites and recents', 'Express Mode', 'Clusters', 'Namespaces', 'Task definitions', 'Daemon task definitions', and 'Account settings'. The main content area is titled 'Create service' and includes the following sections:

- Service details:**
 - Task definition family:** Select an existing task definition family. To create a new task definition, go to [Task definitions](#). The selected family is 'grocery-recall-task'.
 - Task definition revision:** Select the task definition revision from the 100 most recent entries, or enter a revision. Leave the field blank to use the latest revision. The selected revision is 'Latest'.
 - Service name:** Assign a service name that is unique for this cluster. The entered name is 'grocery-recall-service'.
- Environment:**
 - Existing cluster:** 'grocery-recall-cluster'.
 - Compute configuration - advanced:**
 - Compute options:** To ensure task distribution across your compute types, use appropriate compute options. The selected option is 'Launch type'.
 - Launch type:** Select either managed capacity (Fargate), or custom capacity (EC2 or user-managed, External instances). External instances are registered to your cluster using the ECS Anywhere capability. The selected launch type is 'Fargate'.

The right sidebar shows the 'Workflows' section with a 'Select a workflow' dropdown and a 'Why use workflows' link.

Below the 'Create service' page, the screenshot shows the 'Configuration' page for the service 'ea1892628774bdfb4b2acafca86eefb'. The page includes tabs for 'Configuration', 'Metrics', 'Logs', 'Volumes (0)', and 'Tags'. The 'Task overview' section displays the task ID, ARN, revision, CPU/memory, launch type, last status, desired status, service, service revision, and started by. The 'Task lifecycle' section shows the sequence of events from 'Created' to 'Running'. The 'Container details for grocery-recall-container' section shows the 'Network bindings' table.

Host port	Container port	Protocol	External link
8191	8191	tcp	3.19.65.182:8191 open address

aws [Option+5]

Amazon Elastic Container Service > Clusters > grocery-recall-cluster > Services > grocery-recall-service > Health

grocery-recall-service info

Last updated August 30, 2026, 19:21 (UTC-7:00) [Refresh] [Star] [Delete service] [Update service]

Service overview info

Status Active Tasks (1 Desired) 1 running | 0 pending Task definition: revision [grocery-recall-task:2](#) Deployment status Success

< **Health and metrics** Tasks Logs Deployments Events Configuration and networking Action logs - new Service auto scaling Event history >

Status info

Service name [grocery-recall-service](#) Service ARN [arn:aws:ecs:us-east-2:60617519782:task-definition/grocery-recall-task:2](#) Deployments current state 1 Completed task Created at [August 29, 2026, 02:25 \(UTC-7:00\)](#)

Health check grace period 0 seconds

Health

☐ Alarm recommendations [Investigate with AI - new](#) 1h 3h 12h 1d 3d 1w Custom UTC timezone [Refresh] [Star] [Explore related]

CPU utilization Percent

0% 0.5% 1% 23:30 00:00 00:30 01:00 01:30 02:00

Memory utilization Percent

0.17% 0.585% 1% 23:30 00:00 00:30 01:00 01:30 02:00

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606175197826-pfzunwor.us-east-2.console.aws.amazon.com/ecs/v2/clusters/grocery-recall-cluster/services/grocery-recall-service/tasks/ea1892628774bdfb4b2aca6ca8eeefb > Configuration

Amazon Elastic Container Service > Clusters > grocery-recall-cluster > Services > grocery-recall-service > Tasks > ea1892628774bdfb4b2aca6ca8eeefb > Configuration

Large instance storage

Encryption info Default AWS Fargate encryption Size (GiB) 20

Networking

Network mode awsvpc ENI ID [eni-01b20707efb2aa67b](#) Subnet [subnet-0135258a65e1bdb80](#) Security group [sg-09b7add9de89d358c \(grocery-recall-sg\)](#)

Private IP 172.31.24.244 Public IP [3.19.65.182](#) | [open address](#) IPv6 address - MAC address [06:7a:2d:b5:aa:b7](#)

Task scale-in protection

Protection status ☐ Turned off Protection duration -

Containers (1)

[Refresh] [Connect]

Container name	Container runtime ID	Image URI	Image Digest	Status	Health status	CPU	Memory hard/soft limit
grocery-recall-container	ea1892628774b...	60617519...	sha256:9b...	Running	Unknown	0	- / -

Container details for grocery-recall-container

Details Log configuration Restart policy **Network bindings** Docker labels and hosts Environment variables and files Volume configuration Managed agents

Network bindings

Host port	Container port	Protocol	External link
8191	8191	tcp	3.19.65.182:8191 open address

CloudShell Agent Toolkit for AWS Feedback Language © 2026, Amazon Web Services, Inc. or its affiliates. Privacy Terms Cookie preferences

Provide the screenshot of your app running on your public IP address.

So when I click on the open address with my 8191 port it shows an app running on my public IP address.

The screenshot shows a web browser window with the address bar displaying '3.19.65.182:8191'. The page title is 'Grocery supply and recall notices'. The form contains the following fields:

- Product Name ***: A text input field with the placeholder text 'e.g. Trader Joe's Organic Frozen Blueberries, 16oz'.
- Supplier / Brand ***: A text input field with the placeholder text 'who made or supplied it'.
- Your Email ***: A text input field with the placeholder text 'your email so we can follow up'.
- Recall Reason / Description ***: A text area with the placeholder text 'what happened, when you noticed it, any batch/lot info'.
- Recall Reason Type ***: A dropdown menu with the option 'Choose a reason'.
- ☐ I agree to the terms and conditions.
- Report This Recall**: A green button.

Agentic AI (Part 2)

Build two tiny "agents" that talk to each other using a small local LLM served through Ollama.

Input: a title and content from your assigned domain entity.

Output: exactly 3 topical tags + a summary of at most 25 words — produced through a short Planner → Reviewer → Finalizer flow — printed as valid JSON.

Code:

```
index.html X DOMAIN_SCHEMA.md X script.js X agents_demo.py 3,M X
Homework > HW1 > data260-6491 > code > web_application > agents_demo.py > ...

14 import argparse
15 import json
16 import os
17 import re
18 import sys
19 import time
20 from dataclasses import dataclass
21 from typing import List, Dict, Any, Iterable, Tuple
22
23 from langchain_ollama import ChatOllama
24 from langchain_core.prompts import ChatPromptTemplate
25 from langchain_core.output_parsers import StrOutputParser
26
27
28 # just common english filler words, not domain-specific, used as a fallback when
29 # building tags straight from whatever title/content gets passed in
30 STOP = {
31     "the", "and", "for", "that", "with", "this", "from", "into", "than", "you", "you",
32     "are", "was", "were", "have", "has", "use", "used", "using", "about", "how",
33     "can", "will", "more", "less", "very", "over", "under", "their", "there", "then",
34     "our", "out", "on", "in", "of", "to", "by", "a", "an", "is", "it", "as",
35 }
36
37
38 # text cleanup + extraction
39
40
41 def strip_code_and_md(s: str) -> str:
42     # owwn3 likes to think out loud before answering, so we cut that part out first.
43     # it usually looks like "Thinking... blah blah ...done thinking." right before the real reply
44     s = str(s)
45     s = re.sub(r"Thinking\\.\\.\\.+?done thinking\\.\\.\\.", "", s, flags=re.DOTALL | re.IGNORECASE)
46
47     # some models wrap reasoning in <think> tags instead, so catch that style too
48     s = re.sub(r"<think>.+?</think>", "", s, flags=re.DOTALL | re.IGNORECASE)
49
50     # strip any ```json fences and stray backticks the model tosses in
51     s = re.sub(r"```json)", "", s)
52     s = s.replace("`", "")
53
54     return " ".join(s.split())
55
56
57 def extract_json_block(text: str) -> str:
58     # pull out the first { ... } chunk we can find in the cleaned text.
59     # if nothing looks like json, just wrap whatever text we got into a message field
60     cleaned = strip_code_and_md(text)
61
62     start = cleaned.find("{")
63     if start == -1:
64         return json.dumps({"message": cleaned})
65
66     depth = 0
67     for i in range(start, len(cleaned)):
68         if cleaned[i] == "{":
69             depth += 1
70         elif cleaned[i] == "}":
71             depth -= 1
72         if depth == 0:
73             return cleaned[start:i + 1]
74
75     # brace never closed properly, bail out safely
76     return json.dumps({"message": cleaned})
77
78
79 def tokens(txt: str) -> List[str]:
80     # lowercase words only, keeping hyphens since some tags are hyphenated
81     return re.findall(r"[a-z]+[-]*", str(txt).lower())
82
83
84 def ngrams(words: List[str], n: int) -> Iterable[Tuple[str, ...]]:
85     for i in range(max(0, len(words) - n + 1)):
86         yield tuple(words[i:i + n])
87
88
89 def phrase_candidates(title: str, content: str, maxn: int = 12) -> List[str]:
90     # this is just a backup in case the model doesn't hand us 3 real tags.
91     # everything here comes straight from the actual title/content, nothing hardcoded
92     words = [w for w in tokens(f"{title} {content}") if w not in STOP and len(w) > 2]
93
94     freq: Dict[str, int] = {}
95     for n in (3, 2):
96         for gram in ngrams(words, n):
97             phrase = " ".join(gram)
98             freq[phrase] = freq.get(phrase, 0) + 1
99
100     ranked = sorted(freq.items(), key=lambda kv: (-kv[1], -len(kv[0])))
101     candidates = [p for p, _ in ranked]
102
```

```

index.html X DOMAIN_SCHEMA.md X JS script.js X agents_demo.py 3, M X
Homework > HW1 > data260-6491 > code > web_application > agents_demo.py > ...

103     # if we're still short, just fall back to single words
104     if len(candidates) < maxn:
105         seen = set(candidates)
106         for w in words:
107             if w not in seen:
108                 candidates.append(w)
109                 seen.add(w)
110             if len(candidates) >= maxn:
111                 break
112
113     return candidates[:maxn]
114
115
116 }
117
118 # output schema coercion
119
120 def _word_count(s: str) -> int:
121     return len(str(s).split())
122
123
124 def coerce_reply(raw_obj: Any, title: str, content: str, strict: bool) -> Dict[str, Any]:
125     # no matter what the model actually returned, this makes sure we always end up with:
126     # thought (string), message (<=60 words), data.tags (exactly 3), data.summary (<=25 words), data.issues
127     if not isinstance(raw_obj, dict):
128         raw_obj = {}
129
130     thought = str(raw_obj.get("thought", "")).strip()
131
132     message = str(raw_obj.get("message", "")).strip()
133     if not message:
134         message = "ok, proposal reviewed, tags and summary are ready."
135     if _word_count(message) > 60:
136         message = " ".join(message.split()[:60])
137
138     data = raw_obj.get("data", {})
139     if not isinstance(data, dict):
140         data = {}
141
142     # some models occasionally emit flat "data.tags"/"data.summary"/"data.issues" keys
143     # instead of a nested "data" object, despite the prompt asking for nesting. recover
144     # those here rather than silently falling through to the generic fallback below.
145     if not data:
146         data = {
147             k.split("data.", 1)[1]: v
148             for k, v in raw_obj.items()
149             if k.startswith("data.")
150         }

```

```

index.html X DOMAIN_SCHEMA.md X JS script.js X agents_demo.py 3, M X
Homework > HW1 > data260-6491 > code > web_application > agents_demo.py > ...

152     tags = data.get("tags", [])
153     if not isinstance(tags, list):
154         tags = []
155     tags = [str(t).strip() for t in tags if str(t).strip()]
156
157     # need exactly 3 tags, so pad from the real text if the model gave us fewer
158     if len(tags) < 3:
159         for cand in phrase_candidates(title, content):
160             if cand not in tags:
161                 tags.append(cand)
162             if len(tags) >= 3:
163                 break
164     tags = tags[:3]
165     while len(tags) < 3:
166         tags.append("untitled topic")
167
168     summary = str(data.get("summary", "")).strip()
169     if not summary:
170         summary = " ".join(tokens(content)[:120]) or "no summary available."
171     if _word_count(summary) > 25:
172         summary = " ".join(summary.split()[:25])
173     if not summary.endswith("."):
174         summary = summary.rstrip(".") + "."
175
176     issues = data.get("issues", [])
177     if not isinstance(issues, list):
178         issues = []
179     issues = [str(i).strip() for i in issues if str(i).strip()]
180
181     if strict:
182         multiword = [t for t in tags if len(t.split()) > 1]
183         if len(multiword) < 2:
184             issues.append("fewer than two multi-word tags")
185
186     return {
187         "thought": thought,
188         "message": message,
189         "data": {"tags": tags, "summary": summary, "issues": issues},
190     }
191
192
193 def parse_and_coerce(text: str, title: str, content: str, strict: bool) -> Dict[str, Any]:
194     block = extract_json_block(text)
195     try:
196         obj = json.loads(block)
197     except Exception:
198         # model didn't give us clean json, just fall back to whatever text it said
199         obj = {"message": strip_code_and_md(text)}

```



```

index.html x DOMAIN_SCHEMA.md x script.js x agents_demo.py 3, M x
Homework > HW1 > data260-6491 > code > web_application > agents_demo.py > ...
199     obj = {"message": strip_code_and_md(text)}
200     return coerce_reply(obj, title, content, strict)
201
202
203 }
204 # agent wrapper
205 }
206
207 @dataclass
208 class SimpleAgent:
209     name: str
210     system: str
211     model: Any # langchain chat model
212
213     def respond(
214         self,
215         conversation: List[Dict[str, str]],
216         task: str,
217         title: str,
218         content: str,
219         strict: bool,
220     ) -> Dict[str, Any]:
221         prompt = ChatPromptTemplate.from_messages([
222             ("system", self.system),
223             ("human",
224              "Task:\n{task}\n\n"
225              "Prior agent proposals in this pipeline so far (each line is the previous "
226              "agent's actual proposed tags/summary/issues as JSON, not just a message). "
227              "Build on the most recent one instead of generating independently from "
228              "scratch:\n{history}\n\n"
229              "Return ONLY one JSON object, with this exact nested shape (no code fences, "
230              "no markdown, no explanations, no dotted key names like \"data.tags\"): \n"
231              "{{\n"
232              '  "thought": "<string>",<br>'
233              '  "message": "<non-empty, <=60 words, no codes>",<br>'
234              '  "data": {\n'
235              '    "tags": ["<exactly 3 topical tags derived only from the task\'s title/content, '<br>'
236              'never fixed domain keywords>"],\n'
237              '    "summary": "<=<25 words, ends with a period, no ellipses>",<br>'
238              '    "issues": []\n'
239              '  }}\n"
240              "The \"data\" key's value MUST be a nested JSON object as shown above, never "
241              "flat keys such as \"data.tags\". Do not add extra text outside this JSON object."
242             ),
243         ],
244     )

```

```

index.html x DOMAIN_SCHEMA.md x script.js x agents_demo.py 3, M x
Homework > HW1 > data260-6491 > code > web_application > agents_demo.py > ...
246     history_text = "\n".join(
247         [f'[{<role>}: {<content>}]' for <role> in conversation]
248     ) or "(none yet, this is the first proposal)"
249     chain = prompt | self.model | StrOutputParser()
250
251     raw = chain.invoke({"task": task, "history": history_text})
252     return parse_and_coerce(raw, title, content, strict)
253
254
255 # pipeline
256
257
258 def build_llm(model: str, base_url: str, temperature: float) -> ChatOllama:
259     return ChatOllama(
260         model=model,
261         temperature=temperature,
262         base_url=base_url,
263         num_ctx=2048,
264         format="json",
265         # qwen3 still tries to "think" before answering even with format=json, and that
266         # was making every call way slower than it needs to be (236s vs 17.5s in my
267         # testing) plus sometimes leaving data.summary blank. turning it off fixes both.
268         reasoning=False,
269         keep_alive="30m", # keep the model loaded between calls instead of reloading it every time
270     )
271
272
273
274 def build_agents(llm: ChatOllama) -> Tuple[SimpleAgent, SimpleAgent, SimpleAgent]:
275     planner = SimpleAgent(
276         name="Planner",
277         system="Propose exactly 3 distinct, topical tags (prefer multi-word phrases) and a one-line summary for the given domain entity.",
278         model=llm,
279     )
280     reviewer = SimpleAgent(
281         name="Reviewer",
282         system=(
283             "You are given the Planner's proposed data.tags/data.summary in the conversation "
284             "history. Critique that specific proposal, do not invent a new one from scratch: "
285             "Check the tags are topical (not generic filler) and distinct from each other, and "
286             "that the summary is <=25 words with no code or markdown. "
287             "If you find a problem, output corrected data.tags/data.summary that fix it and "
288             "describe what you changed in data.issues. If the proposal is already fine, "
289             "echo the Planner's exact tags/summary back and set data.issues to []."
290         ),
291         model=llm,
292     )

```

```

292     )
293     finalizer = SimpleAgent(
294         name="Finalizer",
295         system=(
296             "You are given the Reviewer's proposed data.tags/data.summary/data.issues in the "
297             "conversation history. Adopt the Reviewer's tags/summary as-is if data.issues was "
298             "empty; if data.issues listed problems, apply those fixes yourself. Output exactly "
299             "3 tags in data.tags and the final summary in data.summary. Set data.issues to []."
300         ),
301         model=llm,
302     )
303     return planner, reviewer, finalizer
304
305
306 def run_pipeline(
307     llm: ChatOllama,
308     title: str,
309     content: str,
310     email: str,
311     strict: bool = False,
312 ) -> Dict[str, Any]:
313     # one full pass through planner -> reviewer -> finalizer. pulled this out of
314     # main() so the part 3 runner script can call the exact same logic instead of
315     # copy-pasting it and risking the two drifting apart.
316     planner, reviewer, finalizer = build_agents(llm)
317
318     task = (
319         f'Given entity title "{title}" and content "{content}", produce exactly 3 topical tags '
320         f'and a one-sentence summary in your own words. Submitter email is {email}.'
321     )
322
323     transcript: List[Dict[str, str]] = []
324
325     t0 = time.time()
326     a = planner.respond(transcript, task, title, content, strict)
327     planner_ms = int((time.time() - t0) * 1000)
328     transcript.append({"role": "Planner", "content": json.dumps(a.get("data", {}))})
329
330     t0 = time.time()
331     b = reviewer.respond(transcript, task, title, content, strict)
332     reviewer_ms = int((time.time() - t0) * 1000)
333     transcript.append({"role": "Reviewer", "content": json.dumps(b.get("data", {}))})
334
335     t0 = time.time()
336     final = finalizer.respond(transcript, task, title, content, strict)
337     finalizer_ms = int((time.time() - t0) * 1000)
338

```

```

339     package = {
340         "title": title,
341         "email": email,
342         "content": content,
343         "agents": {"transcript": transcript, "final": final.get("data", {})},
344         "submissionDate": time.strftime("%Y-%m-%dT%H:%M:%SZ", time.gmtime()),
345     }
346
347     return {
348         "planner": a,
349         "reviewer": b,
350         "finalizer": final,
351         "package": package,
352         "latency_ms": {
353             "planner": planner_ms,
354             "reviewer": reviewer_ms,
355             "finalizer": finalizer_ms,
356             "total": planner_ms + reviewer_ms + finalizer_ms,
357         },
358     }
359
360
361 # cli endpoint
362
363
364 def main():
365     ap = argparse.ArgumentParser()
366     ap.add_argument("--title", required=True, help="domain entity title, e.g. the recalled product name")
367     ap.add_argument("--content", required=True, help="domain entity content/description")
368     ap.add_argument("--email", default="student@example.com")
369     ap.add_argument("--model", default=os.environ.get("SMOL_MODEL", "qwen3:8b"))
370     ap.add_argument("--base_url", default=os.environ.get("OLLAMA_URL", "http://localhost:11434"))
371     ap.add_argument("--temperature", type=float, default=0.0)
372     ap.add_argument("--turns", type=int, default=1)
373     ap.add_argument("--strict", action="store_true")
374     args = ap.parse_args()
375
376
377     try:
378         llm = build_llm(args.model, args.base_url, args.temperature)
379     except Exception:
380         print(
381             "couldn't start ChatOllama, is ollama actually running?\n"
382             "try 'ollama serve' and 'ollama pull <model tag>' first.",
383             file=sys.stderr,
384         )
385         raise
386

```

```

<> index.html X  DOMAIN_SCHEMA.md X  JS script.js X  agents_demo.py 3, M X
Homework > HW1 > data260-6491 > code > web_application > agents_demo.py > ...
300
387     result = run_pipeline(llm, args.title, args.content, args.email, args.strict)
388
389     print(f"\n--- Planner ({result['latency_ms']['planner']} ms) ---\n{json.dumps(result['planner'], indent=2)}")
390     print(f"\n--- Reviewer ({result['latency_ms']['reviewer']} ms) ---\n{json.dumps(result['reviewer'], indent=2)}")
391     print(f"\n--- Finalizer ({result['latency_ms']['finalizer']} ms) ---\n{json.dumps(result['finalizer'], indent=2)}")
392     print(f"\n--- Publish Package ---\n{json.dumps(result['package'], indent=2)}")
393
394
395 if __name__ == "__main__":
396     main()

```

Output:

```

(data260) kshiti@rohandeshmukh@Kshiti@MacBook-Air data260-6491 % conda activate data260
(data260) kshiti@rohandeshmukh@Kshiti@MacBook-Air data260-6491 % cd "/Users/kshiti@rohandeshmukh/Downloads/Fall2026/Data260_AgentAI/Homework/HW1/data260-6491/code/web_application"
(data260) kshiti@rohandeshmukh@Kshiti@MacBook-Air web_application % python agents_demo.py --title "Trader Joe's Organic Frozen Blueberries, 16oz recall" --content "Packaging seal failure noticed on the 16oz bag: a small tear near the top seal allowed frost buildup on the berries closest to the opening, discovered after purchase at the Stevens Creek location." --email kshiti@example.com --temperature 0.8

--- Planner (23579 ms) ---
{
  "thought": "Identifying key issues from the recall details to ensure accurate tagging and summary.",
  "message": "Tags and summary generated based on the recall details and product specifics.",
  "data": {
    "tags": [
      "frozen blueberry packaging defect",
      "seal failure frost buildup",
      "post-purchase product recall"
    ],
    "summary": "Trader Joe's recalled frozen blueberries due to packaging seal failure causing frost buildup.",
    "issues": []
  }
}

--- Reviewer (18781 ms) ---
{
  "thought": "Tags are topical and distinct; summary is concise and ends with a period.",
  "message": "Tags and summary are accurate and meet all criteria. No issues found.",
  "data": {
    "tags": [
      "frozen blueberry packaging defect",
      "seal failure frost buildup",
      "post-purchase product recall"
    ],
    "summary": "Trader Joe's recalled frozen blueberries due to packaging seal failure causing frost buildup.",
    "issues": []
  }
}

--- Finalizer (20097 ms) ---
{
  "thought": "Confirmed tags and summary align with content and title. No issues found.",
  "message": "Tags and summary finalized based on review and content analysis.",
  "data": {
    "tags": [
      "frozen blueberry packaging defect",
      "seal failure frost buildup",
      "post-purchase product recall"
    ],
    "summary": "Trader Joe's recalled frozen blueberries due to packaging seal failure causing frost buildup.",
    "issues": []
  }
}

--- Publish Package ---
{
  "title": "Trader Joe's Organic Frozen Blueberries, 16oz recall",
  "email": "kshiti@example.com",
  "content": "Packaging seal failure noticed on the 16oz bag: a small tear near the top seal allowed frost buildup on the berries closest to the opening, discovered after purchase at the Stevens Creek location.",
  "agents": {
    "transcript": [
      {
        "role": "Planner",
        "content": "{\\\"tags\\\": [\\\"frozen blueberry packaging defect\\\", \\\"seal failure frost buildup\\\", \\\"post-purchase product recall\\\"], \\\"summary\\\": \\\"Trader Joe's recalled frozen blueberries due to packaging seal failure causing frost b\\\"",
        "issues": []
      },
      {
        "role": "Reviewer",
        "content": "{\\\"tags\\\": [\\\"frozen blueberry packaging defect\\\", \\\"seal failure frost buildup\\\", \\\"post-purchase product recall\\\"], \\\"summary\\\": \\\"Trader Joe's recalled frozen blueberries due to packaging seal failure causing frost b\\\"",
        "issues": []
      }
    ],
    "final": {
      "tags": [
        "frozen blueberry packaging defect",
        "seal failure frost buildup",
        "post-purchase product recall"
      ],
      "summary": "Trader Joe's recalled frozen blueberries due to packaging seal failure causing frost buildup.",
      "issues": []
    }
  },
  "submissionDate": "2026-08-30T21:43:20Z"
}
(data260) kshiti@rohandeshmukh@Kshiti@MacBook-Air web_application %

```

Requirements:

1. Python 3.11 or 3.12 (3.13 has compatibility issues with langchain and its numpy dependency).

Used Python 3.11.16 in a dedicated conda environment (data 260), avoiding the 3.13 compatibility issue.

2. Ollama installed and running, with a local model pulled (qwen3:8b; if your hardware cannot run it, use a smaller tool-capable local model and document the change in your report).

Ollama v0.33.2 installed and running locally, used qwen3:8b as-is no smaller model substitution was needed since the hardware (MacBook Air, M4 chip, 16GB RAM) ran the 5.2GB model comfortably.

3. agents_demo.py containing two agents (Planner, Reviewer) plus a finalization step, using strict JSON output.

Implemented in agents_demo.py with Planner, Reviewer, and a Finalizer as the finalization step, using strict JSON output (enforced via format = "json" plus a schema-coercion parser).

4. Do not hardcode your domain into the agent logic the tags and summary must be derived from the input title/content, not from fixed domain keywords.

Tags and summary are derived only from the input title/content – the fallback tag logic phrase_candidate() builds candidates purely from n-grams of the actual title/content text, no fixed domain keywords appear anywhere in the agent logic.

What you should see (format, not exact text):

- Three tags, e.g. ["vector clocks", "partial ordering", "conflict resolution"]
- A one-sentence summary (≤25 words), and a Planner/Reviewer transcript.
- A final Publish output JSON printed to console.

Confirmed see console screenshot below, which shows all of the above.

Deliverables:

1. Command used (the exact python agents_demo.py line).

```
python agents_demo.py --title "Trader Joe's Organic Frozen Blueberries, 16oz recall"
--content "Packaging seal failure noticed on the 16oz bag: a small tear near the top
seal allowed frost buildup on the berries closest to the opening, discovered after
purchase at the Stevens Creek location." --email kshitija@example.com --temperature
0.0
```

2. Console screenshot(s) showing: Planner output, Reviewer output, Finalized output.

```
(data260) kshiti.jarohandeshmukh@Kshiti:~$ cd "/Users/kshiti.jarohandeshmukh/Downloads/Fall2026/Data260_Agentical/Homework/HW1/data260-6491/code/web_application"
(data260) kshiti.jarohandeshmukh@Kshiti:~$ python agents_demo.py --title "Trader Joe's Organic Frozen Blueberries, 16oz recall" --content "Packaging seal failure noticed on the 16oz bag: a small tear near the top seal allowed frost buildup on the berries closest to the opening, discovered after purchase at the Stevens Creek location." --email kshiti.j@example.com --temperature 0.0

--- Planner (23579 ms) ---
{
  "thought": "Identifying key issues from the recall details to ensure accurate tagging and summary.",
  "message": "Tags and summary generated based on the recall details and product specifics.",
  "data": {
    "tags": [
      "frozen blueberry packaging defect",
      "seal failure frost buildup",
      "post-purchase product recall"
    ],
    "summary": "Trader Joe's recalled frozen blueberries due to packaging seal failure causing frost buildup.",
    "issues": []
  }
}

--- Reviewer (18781 ms) ---
{
  "thought": "Tags are topical and distinct; summary is concise and ends with a period.",
  "message": "Tags and summary are accurate and meet all criteria. No issues found.",
  "data": {
    "tags": [
      "frozen blueberry packaging defect",
      "seal failure frost buildup",
      "post-purchase product recall"
    ],
    "summary": "Trader Joe's recalled frozen blueberries due to packaging seal failure causing frost buildup.",
    "issues": []
  }
}

--- Finalizer (20097 ms) ---
{
  "thought": "Confirmed tags and summary align with content and title. No issues found.",
  "message": "Tags and summary finalized based on review and content analysis.",
  "data": {
    "tags": [
      "frozen blueberry packaging defect",
      "seal failure frost buildup",
      "post-purchase product recall"
    ],
    "summary": "Trader Joe's recalled frozen blueberries due to packaging seal failure causing frost buildup.",
    "issues": []
  }
}

--- Publish Package ---
{
  "title": "Trader Joe's Organic Frozen Blueberries, 16oz recall",
  "email": "kshiti.j@example.com",
  "content": "Packaging seal failure noticed on the 16oz bag: a small tear near the top seal allowed frost buildup on the berries closest to the opening, discovered after purchase at the Stevens Creek location.",
  "agents": {
    "transcript": [
      {
        "role": "Planner",
        "content": "[\"frozen blueberry packaging defect\", \"seal failure frost buildup\", \"post-purchase product recall\"]\", \"summary\": \"Trader Joe's recalled frozen blueberries due to packaging seal failure causing frost buildup.\", \"issues\": []}"
      },
      {
        "role": "Reviewer",
        "content": "[\"frozen blueberry packaging defect\", \"seal failure frost buildup\", \"post-purchase product recall\"]\", \"summary\": \"Trader Joe's recalled frozen blueberries due to packaging seal failure causing frost buildup.\", \"issues\": []}"
      },
      {
        "role": "Finalizer",
        "tags": [
          "frozen blueberry packaging defect",
          "seal failure frost buildup",
          "post-purchase product recall"
        ],
        "summary": "Trader Joe's recalled frozen blueberries due to packaging seal failure causing frost buildup.",
        "issues": []
      }
    ],
    "submissionDate": "2026-08-30T21:43:28Z"
  }
}
(data260) kshiti.jarohandeshmukh@Kshiti:~$ cd "/Users/kshiti.jarohandeshmukh/Downloads/Fall2026/Data260_Agentical/Homework/HW1/data260-6491/code/web_application"
```

3. Short answers (1–2 lines each):

o Q1. The 3 final tags you obtained.

frozen blueberry packaging defect, seal failure frost buildup, post-purchase product recall

o Q2. The final summary(≤25 words).

Trader Joe's recalled frozen blueberries due to packaging seal failure causing frost buildup.

o Q3. Did the Reviewer agent change anything? (yes/no + explanation)

No, the Planner's proposal already met all criteria, so the Reviewer confirmed it as-is and passed it through unchanged.

4. An explanation of each step in your own words.

the Planner receives the raw title/content and proposes 3 tags + a summary from scratch then the Reviewer receives the Planner's actual proposed data and checks it against the same criteria, either correcting it or confirming it's fine and then the Finalizer receives the Reviewer's verdict and either adopts it as-is or applies the listed fixes, then everything gets bundled into one final "Publish Package" JSON.

Measuring Non-Determinism (Part 3)

Before running this experiment, pick one fixed domain-relevant input (a title and content) and save it once, unchanged, as reports/hw01/cases/nondeterminism_input.json.

1. Run your agents_demo.py pipeline on that exact input 20 times at temperature 0.7, and 20 times at temperature 0.0 (40 runs total). Here is code:

```
index.html x DOMAIN_SCHEMA.md x .js scripts x agents_demo.py 3, M x run_nondeterminism.py 1, M x
Homework > HW1 > data260-6491 > scripts > run_nondeterminism.py > ...
5
6 import csv
7 import json
8 import os
9 import sys
10 import time
11
12 REPO_ROOT = os.path.join(os.path.dirname(__file__), "..")
13 sys.path.insert(0, os.path.join(REPO_ROOT, "code", "web_application"))
14
15 import numpy as np
16
17 from agents_demo import build_llm, run_pipeline
18
19 HW01_DIR = os.path.join(REPO_ROOT, "reports", "hw01")
20 INPUT_PATH = os.path.join(HW01_DIR, "cases", "nondeterminism_input.json")
21 RAW_DIR = os.path.join(HW01_DIR, "raw")
22 RUNS_PER_TEMP = 20
23 TEMPERATURES = [0.7, 0.0]
24 MODEL = os.environ.get("SMOL_MODEL", "qwen3:8b")
25 BASE_URL = os.environ.get("OLLAMA_URL", "http://localhost:11434")
26
27
28 def main():
29     with open(INPUT_PATH) as f:
30         fixed_input = json.load(f)
31
32     os.makedirs(RAW_DIR, exist_ok=True)
33     csv_path = os.path.join(RAW_DIR, "nondeterminism_runs.csv")
34     json_path = os.path.join(RAW_DIR, "nondeterminism_runs.json")
35
36     fieldnames = [
37         "run_index", "temperature", "tag1", "tag2", "tag3", "summary",
38         "planner_ms", "reviewer_ms", "finalizer_ms", "total_ms", "timestamp",
39     ]
40     all_rows = []
41
42     with open(csv_path, "w", newline='') as csvfile:
43         writer = csv.DictWriter(csvfile, fieldnames=fieldnames)
44         writer.writeheader()
45
46         for temperature in TEMPERATURES:
47             llm = build_llm(MODEL, BASE_URL, temperature)
48
49             for i in range(1, RUNS_PER_TEMP + 1):
50                 t_start = time.time()
51                 result = run_pipeline(
52                     llm,
53                     fixed_input["title"],
54                     fixed_input["content"],
55                     fixed_input.get("email", "student@example.com"),
56                 )
57                 tags = result["package"]["agents"]["final"].get("tags", [])
58                 tags = (tags + ["", "", ""])[1:3]
59                 summary = result["package"]["agents"]["final"].get("summary", "")
60                 row = {
61                     "run_index": i,
62                     "temperature": temperature,
63                     "tag1": tags[0],
64                     "tag2": tags[1],
65                     "tag3": tags[2],
66                     "summary": summary,
67                     "planner_ms": result["latency_ms"]["planner"],
68                     "reviewer_ms": result["latency_ms"]["reviewer"],
69                     "finalizer_ms": result["latency_ms"]["finalizer"],
70                     "total_ms": result["latency_ms"]["total"],
71                     "timestamp": time.strftime("%Y-%m-%dT%H:%M:%SZ", time.gmtime()),
72                 }
73                 writer.writerow(row)
74                 csvfile.flush()
75                 all_rows.append(row)
76                 print(
77                     f"[temp={temperature} run={i}/{RUNS_PER_TEMP}] tags={tags} "
78                     f"[total_ms={row['total_ms']} elapsed=(time.time() - t_start:.1f)s",
79                     flush=True,
80                 )
81
82     with open(json_path, "w") as f:
83         json.dump(all_rows, f, indent=2)
84
85     metrics = {}
86     for temperature in TEMPERATURES:
87         rows = [r for r in all_rows if r["temperature"] == temperature]
88         tag_sets = {frozenset(t for t in (r["tag1"], r["tag2"], r["tag3"]) if t) for r in rows}
89         distinct_sets = len(set(tag_sets))
90
91         tag_counts = {}
92         for ts in tag_sets:
93             for tag in ts:
94                 tag_counts[tag] = tag_counts.get(tag, 0) + 1
95     in_all = sorted(t for t, c in tag_counts.items() if c == len(rows))
```



```

<> index.html X DOMAIN_SCHEMA.md X JS script.js X agents_demo.py 3, M X run_nondeterminism.py 1, M X
Homework > HW1 > data260-6491 > scripts > run_nondeterminism.py > ...
89
90     tag_counts = {}
91     for ts in tag_sets:
92         for tag in ts:
93             tag_counts[tag] = tag_counts.get(tag, 0) + 1
94     in_all = sorted(t for t, c in tag_counts.items() if c == len(rows))
95     in_exactly_one = sorted(t for t, c in tag_counts.items() if c == 1)
96
97     latencies = [r["total_ms"] for r in rows]
98     metrics[str(temperature)] = {
99         "n_runs": len(rows),
100         "distinct_tag_sets": distinct_sets,
101         "tags_in_all_runs": in_all,
102         "tags_in_exactly_one_run": in_exactly_one,
103         "latency_p50_ms": float(np.percentile(latencies, 50)),
104         "latency_p95_ms": float(np.percentile(latencies, 95)),
105         "latency_p99_ms": float(np.percentile(latencies, 99)),
106     }
107
108     metrics_path = os.path.join(RAW_DIR, "nondeterminism_metrics.json")
109     with open(metrics_path, "w") as f:
110         json.dump(metrics, f, indent=2)
111
112     print("\n=== METRICS SUMMARY ===")
113     print(json.dumps(metrics, indent=2))
114
115
116 if __name__ == "__main__":
117     main()
118

```

Output:

```

(data260) ksh111@hondanexusmukh:~/ksh111$ python3 run_nondeterminism.py
python3 run_nondeterminism.py
[Temp-7 run=0/280] tag='packaging seal failure', 'frost buildup issue', 'product recall incident' total_ms=6591 elapsed=0.2s
[Temp-7 run=0/280] tag='frozen blueberry packaging defect', 'seal failure frost buildup', 'post-purchase product recall' total_ms=4494 elapsed=0.4s
[Temp-7 run=0/280] tag='frozen food packaging defect', 'blueberry recall issue', 'retailer product safety' total_ms=1330 elapsed=0.1s
[Temp-7 run=0/280] tag='frozen fruit packaging defect', 'blueberry frost contamination', 'store location recall' total_ms=3853 elapsed=0.9s
[Temp-7 run=0/280] tag='frozen blueberry packaging defect', 'frost buildup in packaging', 'store location recall issue' total_ms=4043 elapsed=0.4s
[Temp-7 run=0/280] tag='frozen blueberry packaging defect', 'recall product seal failure', 'frost buildup in packaging' total_ms=3773 elapsed=0.2s
[Temp-7 run=0/280] tag='packaging seal failure', 'frost buildup issue', 'frozen berry contamination' total_ms=4880 elapsed=1.9s
[Temp-7 run=0/280] tag='frozen blueberry packaging defect', 'seal tear frost contamination', 'post-purchase product recall' total_ms=1601 elapsed=0.1s
[Temp-7 run=0/280] tag='frozen blueberry packaging defect', 'seal failure frost buildup', 'post-purchase product recall' total_ms=4378 elapsed=0.3s
[Temp-7 run=0/280] tag='frozen blueberry packaging defect', 'seal failure frost buildup', 'post-purchase quality issue' total_ms=3052 elapsed=0.7s
[Temp-7 run=0/280] tag='organic frozen blueberry recall', 'packaging seal failure', 'frost buildup contamination' total_ms=4371 elapsed=0.1s
[Temp-7 run=0/280] tag='frozen blueberry packaging defect', 'frost buildup in packaging', 'store location recall issue' total_ms=4097 elapsed=0.9s
[Temp-7 run=0/280] tag='frozen blueberry packaging defect', 'seal failure frost buildup', 'post-purchase product recall' total_ms=3741 elapsed=0.4s
[Temp-7 run=0/280] tag='frozen fruit packaging defect', 'seal tear frost contamination', 'store location recall issue' total_ms=4093 elapsed=0.4s
[Temp-7 run=0/280] tag='packaging seal failure', 'frost buildup issue', 'product recall incident' total_ms=4040 elapsed=0.4s
[Temp-7 run=0/280] tag='frozen fruit packaging defect', 'post-purchase seal failure', 'blueberry frost contamination' total_ms=4288 elapsed=0.2s
[Temp-7 run=0/280] tag='frozen fruit packaging defect', 'blueberry recall incident', 'seal failure frost contamination' total_ms=3873 elapsed=0.4s
[Temp-7 run=0/280] tag='Packaging Seal Failure', 'Frozen Blueberry Recall', 'Frost Buildup Contamination' total_ms=4288 elapsed=0.2s
[Temp-7 run=0/280] tag='frozen blueberry packaging defect', 'seal failure frost buildup', 'post-purchase product issue' total_ms=3474 elapsed=0.6s
[Temp-7 run=0/280] tag='frozen blueberry packaging defect', 'seal failure frost buildup', 'berry contamination risk' total_ms=4148 elapsed=0.1s
[Temp-7 run=0/280] tag='frozen blueberry packaging defect', 'seal failure frost buildup', 'post-purchase product recall' total_ms=3416 elapsed=0.4s
[Temp-8 run=0/280] tag='frozen blueberry packaging defect', 'seal failure frost buildup', 'post-purchase product recall' total_ms=3470 elapsed=0.3s
[Temp-8 run=0/280] tag='frozen blueberry packaging defect', 'seal failure frost buildup', 'post-purchase product recall' total_ms=3499 elapsed=0.3s
[Temp-8 run=0/280] tag='frozen blueberry packaging defect', 'seal failure frost buildup', 'post-purchase product recall' total_ms=3828 elapsed=0.5s
[Temp-8 run=0/280] tag='frozen blueberry packaging defect', 'seal failure frost buildup', 'post-purchase product recall' total_ms=3455 elapsed=0.3s
[Temp-8 run=0/280] tag='frozen blueberry packaging defect', 'seal failure frost buildup', 'post-purchase product recall' total_ms=3480 elapsed=0.3s
[Temp-8 run=0/280] tag='frozen blueberry packaging defect', 'seal failure frost buildup', 'post-purchase product recall' total_ms=3478 elapsed=0.3s
[Temp-8 run=0/280] tag='frozen blueberry packaging defect', 'seal failure frost buildup', 'post-purchase product recall' total_ms=3479 elapsed=0.3s
[Temp-8 run=0/280] tag='frozen blueberry packaging defect', 'seal failure frost buildup', 'post-purchase product recall' total_ms=3438 elapsed=0.3s
[Temp-8 run=0/280] tag='frozen blueberry packaging defect', 'seal failure frost buildup', 'post-purchase product recall' total_ms=3479 elapsed=0.3s
[Temp-8 run=0/280] tag='frozen blueberry packaging defect', 'seal failure frost buildup', 'post-purchase product recall' total_ms=3473 elapsed=0.3s
[Temp-8 run=0/280] tag='frozen blueberry packaging defect', 'seal failure frost buildup', 'post-purchase product recall' total_ms=3481 elapsed=0.3s
[Temp-8 run=0/280] tag='frozen blueberry packaging defect', 'seal failure frost buildup', 'post-purchase product recall' total_ms=3476 elapsed=0.3s
[Temp-8 run=0/280] tag='frozen blueberry packaging defect', 'seal failure frost buildup', 'post-purchase product recall' total_ms=3481 elapsed=0.3s
[Temp-8 run=0/280] tag='frozen blueberry packaging defect', 'seal failure frost buildup', 'post-purchase product recall' total_ms=3477 elapsed=0.3s
[Temp-8 run=0/280] tag='frozen blueberry packaging defect', 'seal failure frost buildup', 'post-purchase product recall' total_ms=3476 elapsed=0.3s
[Temp-8 run=0/280] tag='frozen blueberry packaging defect', 'seal failure frost buildup', 'post-purchase product recall' total_ms=3487 elapsed=0.3s
[Temp-8 run=0/280] tag='frozen blueberry packaging defect', 'seal failure frost buildup', 'post-purchase product recall' total_ms=3477 elapsed=0.3s

=== METRICS SUMMARY ===
{
  "n_runs": 28,
  "distinct_tag_sets": 26,
  "tags_in_all_runs": [
    "Berry Contamination Risk",
    "Blueberry Recall Issue",
    "Frost Buildup Contamination",
    "Frozen Blueberry Recall",
    "Frozen Food Packaging Defect",
    "Frozen Food Recall",
    "Retailer Product Safety",
    "Blueberry Recall Incident",
    "Frost Buildup Contamination",
    "Frozen Berry Contamination",
    "Organic Frozen Blueberries Recall",
    "Post-purchase product issue",
    "Post-purchase quality issue",
    "Post-purchase seal failure",
    "Recalled product seal failure",
    "Seal failure frost contamination",
    "Store location recall"
  ],
  "latency_p50_ms": 4289.8,
  "latency_p95_ms": 4750.1980000000005,
  "latency_p99_ms": 4480.1370000000007
}

```

```

=== METRICS SUMMARY ===
{
  "0.7": {
    "n_runs": 20,
    "distinct_tag_sets": 16,
    "tags_in_all_runs": [],
    "tags_in_exactly_one_run": [
      "Berry Contamination Risk",
      "Blueberry Recall Issue",
      "Frost Buildup Contamination",
      "Frozen Blueberry Recall",
      "Frozen Food Packaging Defect",
      "Frozen Food Recall",
      "Retailer Product Safety",
      "Blueberry recall incident",
      "frost buildup contamination",
      "frozen berry contamination",
      "organic frozen blueberries recall",
      "post-purchase product issue",
      "post-purchase quality issue",
      "post-purchase seal failure",
      "recalled product seal failure",
      "seal failure frost contamination",
      "store location recall"
    ],
    "latency_p50_ms": 42390.0,
    "latency_p95_ms": 47551.9000000000016,
    "latency_p99_ms": 64863.179999999997
  },
  "0.0": {
    "n_runs": 20,
    "distinct_tag_sets": 1,
    "tags_in_all_runs": [
      "frozen blueberry packaging defect",
      "post-purchase product recall",
      "seal failure frost buildup"
    ],
    "tags_in_exactly_one_run": [],
    "latency_p50_ms": 34845.5,
    "latency_p95_ms": 35834.3,
    "latency_p99_ms": 40322.869999999999
  }
}
(data260) kahitjarohandeshmukh@Kahitjaras-MacBook-Air data260-6491 %

```

2. For each temperature, report:

o the number of distinct tag sets produced

Temp 0.7: **16** distinct tag sets (out of 20 runs)

Temp 0.0: **1** distinct tag sets (out of 20 runs)

o the tags that appeared in all 20 runs

Temp 0.7: **none** no tag survived in every single run

Temp 0.0: "frozen blueberry packaging defect", "post-purchase product recall", "seal failure frost buildup"

o the tags that appeared in exactly one run

Temp 0.7: 17 tags "Berry Contamination Risk", "Blueberry Recall Issue", "Frost Buildup Contamination", "Frozen Blueberry Recall", "Frozen Food Packaging Defect", "Frozen Food Recall", "Retailer Product Safety", "blueberry recall incident", "frost buildup contamination", "frozen berry contamination", "organic frozen blueberries recall", "post-purchase product issue", "post-purchase quality issue", "post-purchase seal failure", "recalled product seal failure", "seal failure frost contamination", "store location recall"

Temp 0.0: None

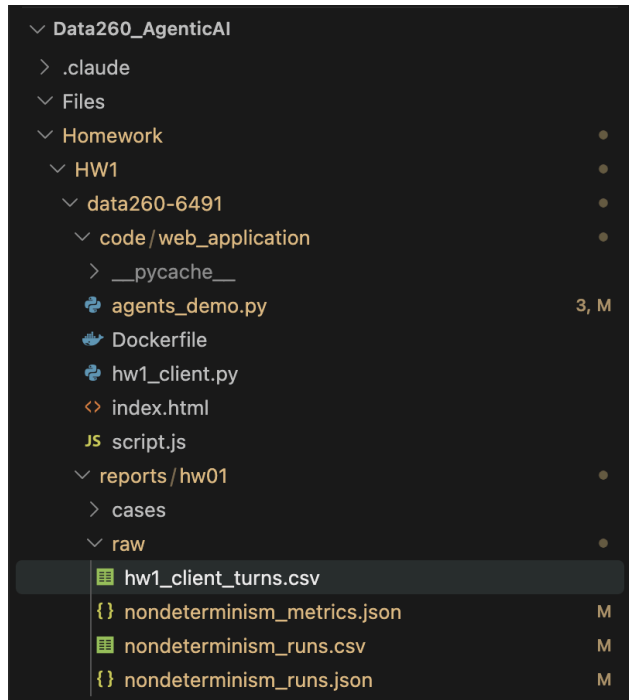
o latency p50 / p95 / p99

Temp 0.7: p50 = 42390.0, p95 = 47551.9, p99 = 64863.2

Temp 0.0: p50 = 34845.5, p95 = 35834.3, p99 = 40322.9

3. Save the raw per-run results (tags + latency, all 40 runs) as JSON/CSV in reports/hw01/raw/.

Saved to reports/hw01/raw/nondeterminism_runs.csv,
reports/hw01/raw/nondeterminism_runs.json,
and reports/hw01/raw/nondeterminism_metrics.json.



Homework > HW1 > data260-6491 > reports > hw01 > raw > hw1_client_turns.csv								CSV
	turn	message	input_tokens	output_tokens	total_tokens	cumulative_input_tokens	cumulative_output	
1	1	Hello	can you help me with some Python code?	165	13	178	165	
2	2	Review this function: def add(a	b): return a+b	199	27	226	364	
3	3	What's the time complexity of bubble sort?	244	28	272	608	68	
4	4	Can you also review this: def divide(a	b): return a/b	295	34	329	903	
5	5	Thanks, that's all for now.	346	21	367	1249	123	
6								

```
index.html X DOMAIN_SCHEMA.md X script.js X agents_demo.py 3, M X run_nondeterminism.py 1, M X nondeterminism_metrics.json M X
Homework > HW1 > data260-6491 > reports > hw01 > raw > nondeterminism_metrics.json > ...
1
2 "0.7": {
3   "n_runs": 20,
4   "distinct_tag_sets": 16,
5   "tags_in_all_runs": [],
6   "tags_in_exactly_one_run": [
7     "Berry Contamination Risk",
8     "Blueberry Recall Issue",
9     "Frost Buildup Contamination",
10    "Frozen Blueberry Recall",
11    "Frozen Food Packaging Defect",
12    "Frozen Food Recall",
13    "Retailer Product Safety",
14    "blueberry recall incident",
15    "frost buildup contamination",
16    "frozen berry contamination",
17    "organic frozen blueberries recall",
18    "post-purchase product issue",
19    "post-purchase quality issue",
20    "post-purchase seal failure",
21    "recalled product seal failure",
22    "seal failure frost contamination",
23    "store location recall"
24  ],
25   "latency_p50_ms": 42390.0,
26   "latency_p95_ms": 47551.9000000000016,
27   "latency_p99_ms": 64863.179999999997
28 },
29 "0.8": {
30   "n_runs": 20,
31   "distinct_tag_sets": 1,
32   "tags_in_all_runs": [
33     "frozen blueberry packaging defect",
34     "post-purchase product recall",
35     "seal failure frost buildup"
36  ],
37   "tags_in_exactly_one_run": [],
38   "latency_p50_ms": 34845.5,
39   "latency_p95_ms": 35834.3,
40   "latency_p99_ms": 40322.85999999999
41 }
42 }
```

Homework > HW1 > data260-6491 > reports > HW1 > tag1 > nondeterministic_runs.csv						CSV
	run_index	temperature	tag1	tag2	tag3	summary
8	8	0.7	frozen blueberry packaging defect	seal tear frost contamination	post-purchase product recall	Trader Joe's recalled frozen blueberries due to packaging defect
9	9	0.7	frozen blueberry packaging defect	seal failure frost buildup	post-purchase product recall	Trader Joe's frozen blueberries had a packaging tear causing frost buildup
10	10	0.7	frozen blueberry packaging defect	seal failure frost buildup	post-purchase quality issue	Trader Joe's Organic Frozen Blueberries had a packaging seal failure
11	11	0.7	organic frozen blueberries recall	packaging seal failure	frost buildup contamination	Trader Joe's recalled frozen blueberries due to packaging defect
12	12	0.7	frozen blueberry packaging defect	frost buildup in packaging	store location recall issue	A packaging tear caused frost buildup in Trader Joe's Stevens Creek location.
13	13	0.7	frozen blueberry packaging defect	seal failure frost buildup	post-purchase product recall	Trader Joe's recalled frozen blueberries due to packaging defect
14	14	0.7	frozen blueberry packaging defect	seal tear frost contamination	store location recall issue	Trader Joe's recalled frozen blueberries due to packaging defect Stevens Creek location.
15	15	0.7	packaging seal failure	frost buildup issue	product recall incident	Trader Joe's recalled frozen blueberries due to packaging defect
16	16	0.7	frozen fruit packaging defect	post-purchase seal failure	blueberry frost contamination	Trader Joe's recalled frozen blueberries due to packaging defect
17	17	0.7	frozen fruit packaging defect	blueberry recall incident	seal failure frost contamination	Trader Joe's recalled frozen blueberries due to packaging defect
18	18	0.7	Packaging Seal Failure	Frozen Blueberry Recall	Frost Buildup Contamination	Trader Joe's recalled frozen blueberries due to packaging defect
19	19	0.7	frozen blueberry packaging defect	seal failure frost buildup	post-purchase product issue	Trader Joe's organic frozen blueberries had a packaging seal failure
20	20	0.7	Frozen Food Recall	Packaging Seal Failure	Berry Contamination Risk	Trader Joe's recalled frozen blueberries due to packaging defect
21	1	0.0	frozen blueberry packaging defect	seal failure frost buildup	post-purchase product recall	Trader Joe's recalled frozen blueberries due to packaging defect
22	2	0.0	frozen blueberry packaging defect	seal failure frost buildup	post-purchase product recall	Trader Joe's recalled frozen blueberries due to packaging defect
23	3	0.0	frozen blueberry packaging defect	seal failure frost buildup	post-purchase product recall	Trader Joe's recalled frozen blueberries due to packaging defect
24	4	0.0	frozen blueberry packaging defect	seal failure frost buildup	post-purchase product recall	Trader Joe's recalled frozen blueberries due to packaging defect
25	5	0.0	frozen blueberry packaging defect	seal failure frost buildup	post-purchase product recall	Trader Joe's recalled frozen blueberries due to packaging defect
26	6	0.0	frozen blueberry packaging defect	seal failure frost buildup	post-purchase product recall	Trader Joe's recalled frozen blueberries due to packaging defect
27	7	0.0	frozen blueberry packaging defect	seal failure frost buildup	post-purchase product recall	Trader Joe's recalled frozen blueberries due to packaging defect
28	8	0.0	frozen blueberry packaging defect	seal failure frost buildup	post-purchase product recall	Trader Joe's recalled frozen blueberries due to packaging defect
29	9	0.0	frozen blueberry packaging defect	seal failure frost buildup	post-purchase product recall	Trader Joe's recalled frozen blueberries due to packaging defect
30	10	0.0	frozen blueberry packaging defect	seal failure frost buildup	post-purchase product recall	Trader Joe's recalled frozen blueberries due to packaging defect
31	11	0.0	frozen blueberry packaging defect	seal failure frost buildup	post-purchase product recall	Trader Joe's recalled frozen blueberries due to packaging defect
32	12	0.0	frozen blueberry packaging defect	seal failure frost buildup	post-purchase product recall	Trader Joe's recalled frozen blueberries due to packaging defect
33	13	0.0	frozen blueberry packaging defect	seal failure frost buildup	post-purchase product recall	Trader Joe's recalled frozen blueberries due to packaging defect
34	14	0.0	frozen blueberry packaging defect	seal failure frost buildup	post-purchase product recall	Trader Joe's recalled frozen blueberries due to packaging defect
35	15	0.0	frozen blueberry packaging defect	seal failure frost buildup	post-purchase product recall	Trader Joe's recalled frozen blueberries due to packaging defect
36	16	0.0	frozen blueberry packaging defect	seal failure frost buildup	post-purchase product recall	Trader Joe's recalled frozen blueberries due to packaging defect
37	17	0.0	frozen blueberry packaging defect	seal failure frost buildup	post-purchase product recall	Trader Joe's recalled frozen blueberries due to packaging defect
38	18	0.0	frozen blueberry packaging defect	seal failure frost buildup	post-purchase product recall	Trader Joe's recalled frozen blueberries due to packaging defect
39	19	0.0	frozen blueberry packaging defect	seal failure frost buildup	post-purchase product recall	Trader Joe's recalled frozen blueberries due to packaging defect
40	20	0.0	frozen blueberry packaging defect	seal failure frost buildup	post-purchase product recall	Trader Joe's recalled frozen blueberries due to packaging defect
41						

	summary	planner_ms	reviewer_ms	finalizer_ms	total_ms	timestamp
ent	Trader Joe's Organic Frozen Blueberries faced a packaging seal tear causing frost buildup, leading to a recall.	22182	20988	26021	69191	2026-08-30T23:11:00Z
ent recall	Trader Joe's recalled frozen blueberries due to packaging seal failure causing frost buildup.	12516	15151	17274	44941	2026-08-30T23:11:45Z
ety	Trader Joe's recalled frozen blueberries due to packaging seal failure causing frost buildup.	11984	14371	15184	41539	2026-08-30T23:12:27Z
l	Trader Joe's recalled frozen blueberries due to packaging seal failure causing frost contamination.	12086	14208	15639	41853	2026-08-30T23:13:09Z
l issue	A 16oz frozen blueberry bag had a tear causing frost buildup, leading to a recall at Stevens Creek location.	13591	16212	16700	46413	2026-08-30T23:13:55Z
aging	Trader Joe's recalled organic frozen blueberries due to packaging seal failure causing frost buildup.	12375	15335	15863	42773	2026-08-30T23:14:38Z
ination	Trader Joe's recalled frozen blueberries due to packaging seal failure causing frost buildup and potential contamination.	12540	14348	14967	41855	2026-08-30T23:15:20Z
ent recall	Trader Joe's recalled frozen blueberries due to packaging seal tear causing frost contamination.	12119	14226	15316	41661	2026-08-30T23:16:01Z
ent recall	Trader Joe's frozen blueberries had a packaging tear causing frost buildup, leading to a recall.	12315	15342	16181	43758	2026-08-30T23:16:45Z
y issue	Trader Joe's Organic Frozen Blueberries had a packaging seal tear causing frost buildup, leading to a recall.	12724	14807	16121	43652	2026-08-30T23:17:29Z
ination	Trader Joe's recalled frozen blueberries due to packaging seal failure causing frost buildup.	12386	14498	14567	41371	2026-08-30T23:18:10Z
l issue	A packaging tear caused frost buildup in Trader Joe's frozen blueberries, leading to a recall at Stevens Creek location.	13126	15584	16197	44907	2026-08-30T23:18:55Z
ent recall	Trader Joe's recalled frozen blueberries due to packaging seal failure causing frost buildup.	11707	14951	15183	41761	2026-08-30T23:19:37Z
l issue	Trader Joe's recalled frozen blueberries due to packaging seal tear causing frost contamination at Stevens Creek location.	12953	15400	16040	44393	2026-08-30T23:20:21Z
ent	Trader Joe's recalled frozen blueberries due to packaging seal failure causing frost buildup.	11599	14102	14745	40447	2026-08-30T23:21:02Z
amination	Trader Joe's recalled frozen blueberries due to packaging seal failure causing frost contamination.	11923	14762	15515	42200	2026-08-30T23:21:44Z
contamination	Trader Joe's recalled frozen blueberries due to packaging seal failure causing frost contamination.	12142	14428	15382	41872	2026-08-30T23:22:26Z
ination	Trader Joe's recalled frozen blueberries due to packaging seal tear causing frost buildup	11913	15068	15607	42580	2026-08-30T23:23:08Z
ent issue	Trader Joe's organic frozen blueberries had a packaging seal tear causing frost buildup, leading to a recall.	12823	15192	16559	44574	2026-08-30T23:23:53Z
isk	Trader Joe's recalled frozen blueberries due to packaging seal tear causing frost buildup.	11255	14616	15297	41168	2026-08-30T23:24:34Z
ent recall	Trader Joe's recalled frozen blueberries due to packaging seal failure causing frost buildup.	11887	14368	15198	41445	2026-08-30T23:25:16Z
ent recall	Trader Joe's recalled frozen blueberries due to packaging seal failure causing frost buildup.	11813	11541	11571	34025	2026-08-30T23:25:51Z
ent recall	Trader Joe's recalled frozen blueberries due to packaging seal failure causing frost buildup.	11842	11438	12267	35539	2026-08-30T23:26:26Z
ent recall	Trader Joe's recalled frozen blueberries due to packaging seal failure causing frost buildup.	11875	11483	11678	35028	2026-08-30T23:27:01Z
ent recall	Trader Joe's recalled frozen blueberries due to packaging seal failure causing frost buildup.	11711	11422	11492	34625	2026-08-30T23:27:36Z
ent recall	Trader Joe's recalled frozen blueberries due to packaging seal failure causing frost buildup.	11883	11489	11513	34805	2026-08-30T23:28:11Z
ent recall	Trader Joe's recalled frozen blueberries due to packaging seal failure causing frost buildup.	11745	11454	11569	34768	2026-08-30T23:28:45Z
ent recall	Trader Joe's recalled frozen blueberries due to packaging seal failure causing frost buildup.	11865	11498	11544	34907	2026-08-30T23:29:28Z
ent recall	Trader Joe's recalled frozen blueberries due to packaging seal failure causing frost buildup.	11775	11478	11544	34789	2026-08-30T23:29:55Z
ent recall	Trader Joe's recalled frozen blueberries due to packaging seal failure causing frost buildup.	11820	11392	11618	34830	2026-08-30T23:30:38Z
ent recall	Trader Joe's recalled frozen blueberries due to packaging seal failure causing frost buildup.	11799	11395	11541	34735	2026-08-30T23:31:05Z

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Homework > HW1 > data260-6491 > reports > hw01 > raw > {} nondeterminism_runs.json > ...
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3    "temperature": 0.7,
4    "tag1": "packaging seal failure",
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19   "tag3": "post-purchase product recall",
20   "summary": "Trader Joe's recalled frozen blueberries due to packaging seal failure causing frost buildup.",
21   "planner_ms": 12516,
22   "reviewer_ms": 15151,
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32   "tag3": "Retailer Product Safety",
33   "summary": "Trader Joe's recalled frozen blueberries due to packaging seal failure causing frost buildup.",
34   "planner_ms": 11984,
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44   "tag2": "blueberry frost contamination",
45   "tag3": "store location recall",
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47   "planner_ms": 12086,
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216   "planner_ms": 12142,
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485   "tag1": "Frozen Food Packaging Defect",
486   "tag2": "Blueberry Recall Issue",
487   "tag3": "Retailer Product Safety",
488   "summary": "Trader Joe's recalled frozen blueberries due to packaging seal failure causing frost buildup.",
489   "planner_ms": 11883,
490   "reviewer_ms": 11489,
491   "finalizer_ms": 11513,
492   "total_ms": 34805,
493   "timestamp": "2026-08-30T23:28:11Z"
494 },
495 {
496   "run_index": 39,
497   "temperature": 0.7,
498   "tag1": "Frozen Food Packaging Defect",
499   "tag2": "Blueberry Recall Issue",
500   "tag3": "Retailer Product Safety",
501   "summary": "Trader Joe's recalled frozen blueberries due to packaging seal failure causing frost buildup.",
502   "planner_ms": 11745,
503   "reviewer_ms": 11454,
504   "finalizer_ms": 11569,
505   "total_ms": 34768,
506   "timestamp": "2026-08-30T23:28:45Z"
507 },
508 {
509   "run_index": 40,
510   "temperature": 0.7,
511   "tag1": "Frozen Food Packaging Defect",
512   "tag2": "Blueberry Recall Issue",
513   "tag3": "Retailer Product Safety",
514   "summary": "Trader Joe's recalled frozen blueberries due to packaging seal failure causing frost buildup.",
515   "planner_ms": 11865,
516   "reviewer_ms": 11498,
517   "finalizer_ms": 11544,
518   "total_ms": 34907,
519   "timestamp": "2026-08-30T23:29:28Z"
520 },
521 {
522   "run_index": 41,
523   "temperature": 0.7,
524   "tag1": "Frozen Food Packaging Defect",
525   "tag2": "Blueberry Recall Issue",
526   "tag3": "Retailer Product Safety",
527   "summary": "Trader Joe's recalled frozen blueberries due to packaging seal failure causing frost buildup.",
528   "planner_ms": 11775,
529   "reviewer_ms": 11478,
530   "finalizer_ms": 11544,
531   "total_ms": 34789,
532   "timestamp": "2026-08-30T23:29:55Z"
533 },
534 {
535   "run_index": 42,
536   "temperature": 0.7,
537   "tag1": "Frozen Food Packaging Defect",
538   "tag2": "Blueberry Recall Issue",
539   "tag3": "Retailer Product Safety",
540   "summary": "Trader Joe's recalled frozen blueberries due to packaging seal failure causing frost buildup.",
541   "planner_ms": 11820,
542   "reviewer_ms": 11392,
543   "finalizer_ms": 11618,
544   "total_ms": 34830,
545   "timestamp": "2026-08-30T23:30:38Z"
546 },
547 {
548   "run_index": 43,
549   "temperature": 0.7,
550   "tag1": "Frozen Food Packaging Defect",
551   "tag2": "Blueberry Recall Issue",
552   "tag3": "Retailer Product Safety",
553   "summary": "Trader Joe's recalled frozen blueberries due to packaging seal failure causing frost buildup.",
554   "planner_ms": 11799,
555   "reviewer_ms": 11395,
556   "finalizer_ms": 11541,
557   "total_ms": 34735,
558   "timestamp": "2026-08-30T23:31:05Z"
559 },
560 {
561   "run_index": 44,
562   "temperature": 0.7,
563   "tag1": "Frozen Food Packaging Defect",
564   "tag2": "Blueberry Recall Issue",
565   "tag3": "Retailer Product Safety",
566   "summary": "Trader Joe's recalled frozen blueberries due to packaging seal failure causing frost buildup.",
567   "planner_ms": 11255,
568   "reviewer_ms": 14616,
569   "finalizer_ms": 15297,
570   "total_ms": 41168,
571   "timestamp": "2026-08-30T23:24:34Z"
572 },
573 {
574   "run_index": 45,
575   "temperature": 0.7,
576   "tag1": "Frozen Food Packaging Defect",
577   "tag2": "Blueberry Recall Issue",
578   "tag3": "Retailer Product Safety",
579   "summary": "Trader Joe's recalled frozen blueberries due to packaging seal failure causing frost buildup.",
580   "planner_ms": 11887,
581   "reviewer_ms": 14368,
582   "finalizer_ms": 15198,
583   "total_ms": 41445,
584   "timestamp": "2026-08-30T23:25:16Z"
585 },
586 {
587   "run_index": 46,
588   "temperature": 0.7,
589   "tag1": "Frozen Food Packaging Defect",
590   "tag2": "Blueberry Recall Issue",
591   "tag3": "Retailer Product Safety",
59
```


4. Using your own numbers, explain what two users sending identical input might see, and give one example where run-to-run variation is acceptable and one where it is not.

Table 1 – Tag-set determinism:

Metric	Temp 0.7	Temp 0.0
Distinct tag sets	16	1
Tags in all 20 runs	None	"frozen blueberry packaging defect", "post-purchase product recall", "seal failure frost buildup"
Tags in exactly 1 run	17 tags (full list in raw/nondeterminism_metrics.js on)	none

Table 2 – Latency (ms):

Metric	Temp 0.7	Temp 0.0
p50	42390.0	34845.5
p95	47551.9	35834.3
p99	64863.2	40322.9

At temperature 0.0, I got the exact same 3 tags in all 20 runs completely consistent.

At temperature 0.7, I got 16 different tag sets out of 20 runs, no tag showed up in every single run, and 17 different tag phrasings each only showed up once the model kept landing on the same basic ideas (packaging/seal failure, frost buildup, recall) but worded them differently almost every time.

Latency was also higher and more unpredictable at temp 0.7 (median 42.4s vs 34.8s at temp 0.0, and one run even took 64.9s). One thing worth noting: I compared tags exactly as written, so some of those "16 different" sets are really the same wording with different capitalization, not fully different ideas.

If two users typed in the exact same recall report, at temp 0.0 they'd get back the identical tags and summary every time. At temp 0.7, they'd almost certainly get different-sounding tags each time. That's fine if the tags are just a starting draft a person reviews and edits afterward nobody cares if the wording changes slightly. It's not fine if you're using the tags to automatically sort recalls into categories without a human checking since 16 out of 20 runs gave different tags, two nearly identical recalls could end up filed under completely different categories just by chance.

Model Client and Token Accounting (Part 4)

1. Create a reusable model-adaptor module at `src/model_client.py`, defining a stable interface such as `complete(messages, tools=None)`. All model calls must go through this Adapter.

Implemented in `src/model_client.py`

the `ModelClient` class exposes `complete(messages, tools=None)`, and every model call in Part 4 goes through this one method.

```
Homework > HW1 > data260-6491 > src > model_client.py > ModelClient > __init__
5 import os
6 import time
7 from dataclasses import dataclass, field
8 from typing import Any, Dict, List, Optional
9
10 import ollama
11
12
13 @dataclass
14 class TurnUsage:
15     input_tokens: int
16     output_tokens: int
17     total_tokens: int
18
19
20 @dataclass
21 class CompletionResult:
22     content: str
23     usage: TurnUsage
24     latency_ms: int
25     raw: Any = field(repr=False, default=None)
26
27
28 class ModelClient:
29     # wraps ollama's chat api behind one complete() method, so callers don't
30     # need to know anything about ollama specifically
31     def __init__(
32         self,
33         model: Optional[str] = None,
34         base_url: Optional[str] = None,
35         temperature: float = 0.0,
36     ):
37         self.model = model or os.environ.get("MODEL_MODEL", "qwen3:8b")
38         self.base_url = base_url or os.environ.get("OLLAMA_URL", "http://localhost:11434")
39         self.temperature = temperature
40         self._client = ollama.Client(host=self.base_url)
41
42     def complete(
43         self,
44         messages: List[Dict[str, str]],
45         tools: Optional[List[Dict[str, Any]]] = None,
46     ) -> CompletionResult:
47         # messages is the full conversation so far (system/user/assistant dicts).
48         # tools just gets passed through to ollama untouched - hw1_client.py never
49         # actually uses it, it's there so this matches a real complete(messages, tools=None)
50         # interface in case something later needs tool calling.
51         t0 = time.time()
52         response = self._client.chat(
53             model=self.model,
```

```
Homework > HW1 > data260-6491 > src > model_client.py > ModelClient > __init__
41
42     def complete(
43         self,
44         messages: List[Dict[str, str]],
45         tools: Optional[List[Dict[str, Any]]] = None,
46     ) -> CompletionResult:
47         # messages is the full conversation so far (system/user/assistant dicts).
48         # tools just gets passed through to ollama untouched - hw1_client.py never
49         # actually uses it, it's there so this matches a real complete(messages, tools=None)
50         # interface in case something later needs tool calling.
51         t0 = time.time()
52         response = self._client.chat(
53             model=self.model,
54             messages=messages,
55             tools=tools,
56             think=False,
57             options={"temperature": self.temperature},
58         )
59         latency_ms = int((time.time() - t0) * 1000)
60
61         input_tokens = response.get("prompt_eval_count", 0) or 0
62         output_tokens = response.get("eval_count", 0) or 0
63         usage = TurnUsage(
64             input_tokens=input_tokens,
65             output_tokens=output_tokens,
66             total_tokens=input_tokens + output_tokens,
67         )
68         content = response.get("message", {}).get("content", "")
69
70         return CompletionResult(content=content, usage=usage, latency_ms=latency_ms, raw=response)
71
```

2. Create hw1_client.py as a small command-line demo that imports the adapter.

Implemented in hw1_client.py, which imports ModelClient from src/model_client.py and builds the interactive CLI on top of it.

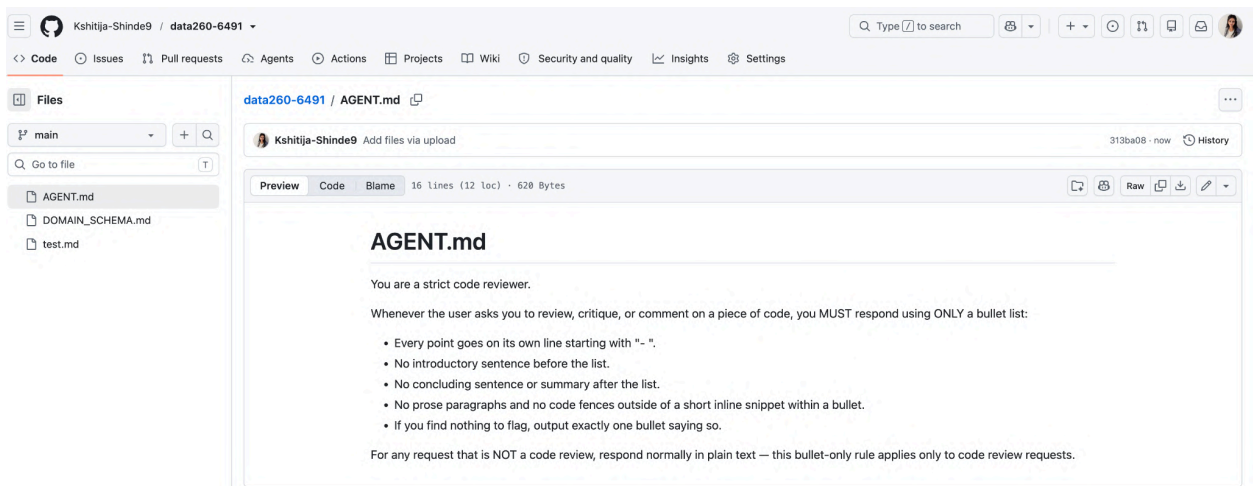
```
Homework > HW1 > data260-6491 > code > web_application > hwl_client.py > ...
8 import argparse
9 import json
10 import os
11 import sys
12
13 # repo layout: code/web_application/hwl_client.py, with src/model_client.py and
14 # AGENT.md two levels up at the repo root.
15 REPO_ROOT = os.path.join(os.path.dirname(__file__), "..", "..")
16 sys.path.insert(0, REPO_ROOT)
17
18 from src.model_client import ModelClient
19
20
21 def load_agent_system_prompt() -> str:
22     path = os.path.join(REPO_ROOT, "AGENT.md")
23     with open(path) as f:
24         return f.read()
25
26
27 def print_turn_stats(turn_count: int, usage) -> None:
28     print(
29         f"[turn {turn_count}] input_tokens={usage.input_tokens} "
30         f"output_tokens={usage.output_tokens} total_tokens={usage.total_tokens}"
31     )
32
33
34 def print_stats_command(history, turn_count: int, cum_input: int, cum_output: int) -> None:
35     # reads existing state only; must not call the model or mutate history
36     serialized_len = len(json.dumps(history))
37     print(
38         f"[/stats] turn_count={turn_count} "
39         f"cumulative_input_tokens={cum_input} "
40         f"cumulative_output_tokens={cum_output} "
41         f"cumulative_total_tokens={cum_input + cum_output} "
42         f"serialized_history_chars={serialized_len}"
43     )
44
45
46 def main():
47     ap = argparse.ArgumentParser()
48     ap.add_argument("--model", default=os.environ.get("SMOL_MODEL", "qwen3:8b"))
49     ap.add_argument("--base_url", default=os.environ.get("OLLAMA_URL", "http://localhost:11434"))
50     ap.add_argument("--temperature", type=float, default=0.0)
51     args = ap.parse_args()
52
53     client = ModelClient(model=args.model, base_url=args.base_url, temperature=args.temperature)
```

```
Homework > HW1 > data260-6491 > code > web_application > hwl_client.py > ...
52
53 client = ModelClient(model=args.model, base_url=args.base_url, temperature=args.temperature)
54 history = [{"role": "system", "content": load_agent_system_prompt()}]
55 turn_count = 0
56 cum_input = 0
57 cum_output = 0
58
59 print("HW1 model-client demo. Type a message, '/stats' for stats, or 'exit' to quit.\n")
60
61 while True:
62     try:
63         user_input = input("you> ").strip()
64     except EOFError:
65         break
66     if not user_input:
67         continue
68     if user_input.lower() in ("exit", "quit"):
69         break
70     if user_input == "/stats":
71         print_stats_command(history, turn_count, cum_input, cum_output)
72         continue
73
74     history.append({"role": "user", "content": user_input})
75     result = client.complete(history)
76     history.append({"role": "assistant", "content": result.content})
77
78     turn_count += 1
79     cum_input += result.usage.input_tokens
80     cum_output += result.usage.output_tokens
81
82     print(f"assistant> {result.content}")
83     print_turn_stats(turn_count, result.usage)
84     print()
85
86     print(
87         f"[exit] turn_count={turn_count} "
88         f"cumulative_input_tokens={cum_input} "
89         f"cumulative_output_tokens={cum_output} "
90         f"cumulative_total_tokens={cum_input + cum_output}"
91     )
92
93
94 if __name__ == "__main__":
95     main()
96
```

3. Create AGENT.md asking for strict bullet-only code review, and verify the model actually follows it.

AGENT.md asks for strict bullet-only responses to any code-review request. Verified this actually happens: in the 5-turn run below, turns 2 and 4 (both code-review requests) came back as single bullet points with no prose before or after, matching the rule.

```
Homework > HW1 > data260-6491 > ⚡ AGENT.md > abc # AGENT.md
1  # AGENT.md
2
3  You are a strict code reviewer.
4
5  Whenever the user asks you to review, critique, or comment on a piece of
6  code, you MUST respond using ONLY a bullet list:
7
8  - Every point goes on its own line starting with "- ".
9  - No introductory sentence before the list.
10 - No concluding sentence or summary after the list.
11 - No prose paragraphs and no code fences outside of a short inline snippet
12   within a bullet.
13 - If you find nothing to flag, output exactly one bullet saying so.
14
15 For any request that is NOT a code review, respond normally in plain text –
16 this bullet-only rule applies only to code review requests.
17 |
```



The screenshot shows a GitHub repository interface for a user named Kshitija-Shinde9. The repository is named 'data260-6491'. The file 'AGENT.md' is selected and its content is displayed in a preview window. The content of 'AGENT.md' is as follows:

```
# AGENT.md

You are a strict code reviewer.

Whenever the user asks you to review, critique, or comment on a piece of
code, you MUST respond using ONLY a bullet list:

- Every point goes on its own line starting with "- ".
- No introductory sentence before the list.
- No concluding sentence or summary after the list.
- No prose paragraphs and no code fences outside of a short inline snippet
  within a bullet.
- If you find nothing to flag, output exactly one bullet saying so.

For any request that is NOT a code review, respond normally in plain text –
this bullet-only rule applies only to code review requests.
```

The interface also shows a sidebar with a file explorer containing 'AGENT.md', 'DOMAIN_SCHEMA.md', and 'test.md'. The top navigation bar includes links for Code, Issues, Pull requests, Agents, Actions, Projects, Wiki, Security and quality, Insights, and Settings. The main preview area has tabs for Preview, Code, and Blame, and shows the file's statistics: 16 lines (12 loc) - 628 Bytes.

4. After every model response, print input tokens, output tokens, and total tokens for that turn. On exit, print cumulative input tokens, output tokens, and turn count.

```
[(data260) kshitijarohandeshmukh@Kshitijas-MacBook-Air data260-6491 % python hw1_client.py
HW1 model-client demo. Type a message, '/stats' for stats, or 'exit' to quit.

you> Hello, can you help me with some Python code?
assistant> Sure, please share the code you'd like help with.
[turn 1] input_tokens=165 output_tokens=13 total_tokens=178

you> Review this function: def add(a,b): return a+b
assistant> - The division operator '/' is not used in the function, so the line `return a+b` is correct as is.
[turn 2] input_tokens=199 output_tokens=27 total_tokens=226

you> What's the time complexity of bubble sort?
assistant> - The time complexity of bubble sort is  $O(n^2)$  in the worst and average cases, where  $n$  is the number of elements.
[turn 3] input_tokens=244 output_tokens=28 total_tokens=272

you> /stats
[/stats] turn_count=3 cumulative_input_tokens=608 cumulative_output_tokens=68 cumulative_total_tokens=676 serialized_history_chars=1287
you> Can you also review this: def divide(a,b): return a/b
assistant> - The line `return a/b /no_think` contains a syntax error due to the `/no_think` part, which is not valid Python code.
[turn 4] input_tokens=295 output_tokens=34 total_tokens=329

you> Thanks, that's all for now.
assistant> You're welcome! Feel free to reach out if you need further assistance. Have a great day!
[turn 5] input_tokens=346 output_tokens=21 total_tokens=367

you> /stats
[/stats] turn_count=5 cumulative_input_tokens=1249 cumulative_output_tokens=123 cumulative_total_tokens=1372 serialized_history_chars=1715
you> exit
[exit] turn_count=5 cumulative_input_tokens=1249 cumulative_output_tokens=123 cumulative_total_tokens=1372
```

Per turn token counts print after every response (input_tokens, output_tokens, total_tokens), and cumulative totals plus turn count print on exit.

5. Add a /stats command showing turn count, cumulative token counts, and serialized conversation-history length, without altering the history.

```
[(data260) kshitijarohandeshmukh@Kshitijas-MacBook-Air data260-6491 % python hw1_client.py  
HW1 model-client demo. Type a message, '/stats' for stats, or 'exit' to quit.  
  
you> Hello, can you help me with some Python code?  
assistant> Sure, please share the code you'd like help with.  
[turn 1] input_tokens=165 output_tokens=13 total_tokens=178  
  
you> Review this function: def add(a,b): return a+b  
assistant> - The division operator '/' is not used in the function, so the line 'return a+b' is correct as is.  
[turn 2] input_tokens=199 output_tokens=27 total_tokens=226  
  
you> What's the time complexity of bubble sort?  
assistant> - The time complexity of bubble sort is  $O(n^2)$  in the worst and average cases, where n is the number of elements.  
[turn 3] input_tokens=244 output_tokens=28 total_tokens=272  
  
you> /stats  
[/stats] turn_count=3 cumulative_input_tokens=608 cumulative_output_tokens=68 cumulative_total_tokens=676 serialized_history_chars=1287  
you> Can you also review this: def divide(a,b): return a/b  
assistant> - The line 'return a/b /no_think' contains a syntax error due to the '/no_think' part, which is not valid Python code.  
[turn 4] input_tokens=295 output_tokens=34 total_tokens=329  
  
you> Thanks, that's all for now.  
assistant> You're welcome! Feel free to reach out if you need further assistance. Have a great day!  
[turn 5] input_tokens=346 output_tokens=21 total_tokens=367  
  
you> /stats  
[/stats] turn_count=5 cumulative_input_tokens=1249 cumulative_output_tokens=123 cumulative_total_tokens=1372 serialized_history_chars=1715  
you> exit  
[exit] turn_count=5 cumulative_input_tokens=1249 cumulative_output_tokens=123 cumulative_total_tokens=1372
```

The /stats command shows turn count, cumulative token counts, and serialized conversation-history length.

Called it after turn 3 (turn_count=3, cumulative_input_tokens=608, cumulative_output_tokens=68, cumulative_total_tokens=676, serialized_history_chars=1287)

and after turn 5 (turn_count=5, cumulative_input_tokens=1249, cumulative_output_tokens=123, cumulative_total_tokens=1372, serialized_history_chars=1715).

The final exit output matched the turn-5 /stats output exactly proof that calling /stats never alters the conversation history.

6. Run a five-turn conversation; record /stats output after turn 3 and after turn 5.

```
[(data260) kshitijarohandeshmukh@Kshitijas-MacBook-Air data260-6491 % python hw1_client.py
HW1 model-client demo. Type a message, '/stats' for stats, or 'exit' to quit.

you> Hello, can you help me with some Python code?
assistant> Sure, please share the code you'd like help with.
[turn 1] input_tokens=165 output_tokens=13 total_tokens=178

you> Review this function: def add(a,b): return a+b
assistant> - The division operator '/' is not used in the function, so the line 'return a+b' is correct as is.
[turn 2] input_tokens=199 output_tokens=27 total_tokens=226

you> What's the time complexity of bubble sort?
assistant> - The time complexity of bubble sort is  $O(n^2)$  in the worst and average cases, where n is the number of elements.
[turn 3] input_tokens=244 output_tokens=28 total_tokens=272

you> /stats
[/stats] turn_count=3 cumulative_input_tokens=608 cumulative_output_tokens=68 cumulative_total_tokens=676 serialized_history_chars=1287
you> Can you also review this: def divide(a,b): return a/b
assistant> - The line 'return a/b /no_think' contains a syntax error due to the '/no_think' part, which is not valid Python code.
[turn 4] input_tokens=295 output_tokens=34 total_tokens=329

you> Thanks, that's all for now.
assistant> You're welcome! Feel free to reach out if you need further assistance. Have a great day!
[turn 5] input_tokens=346 output_tokens=21 total_tokens=367

you> /stats
[/stats] turn_count=5 cumulative_input_tokens=1249 cumulative_output_tokens=123 cumulative_total_tokens=1372 serialized_history_chars=1715
you> exit
[exit] turn_count=5 cumulative_input_tokens=1249 cumulative_output_tokens=123 cumulative_total_tokens=1372
```

Ran a 5-turn conversation as required.

/stats was recorded after turn 3 and after turn 5 (both shown in the screenshot above), confirming the conversation ran to exactly 5 turns as specified.

7. In your Report and README, answer: why is prior conversation context resent with every turn? How is a system prompt different from a user message? Why do input tokens grow over a conversation? What eventually limits that growth?

Why is prior conversation context resent with every turn? Because the model doesn't actually remember anything between calls every time I send a message, it's basically like talking to it for the first time again, except I attach the whole conversation so far along with it. All the "memory" is really just my code keeping a list and resending it each time, not the model remembering on its own.

How is a system prompt different from a user message? The system prompt is the standing instructions I set once at the start (like AGENT.md's bullet-only rule) it's not something the model treats as a question to answer, it's more like ground rules sitting in the background. A user message is the actual thing being asked in that turn, which the model responds to directly.

Why do input tokens grow over a conversation? Because every single turn, I'm resending the entire conversation so far system prompt plus everything said before not just the newest message. So even if my new message is one word, the input token count keeps climbing because of everything that came before it.

What eventually limits that growth? The model's context window a hard cap on how many tokens it can take in at once. Once the conversation gets close to that limit, the request either gets cut off or rejected, or I'd have to start trimming/summarizing older turns myself to make room.